NOVA IMS INFORMATION MANAGEMENT SCHOOL

09

Decision Trees

Data Mining II | MGI 2526

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Agenda

00 Decision Trees

01 Classification Trees

02 Regression Trees

03 How to avoid overfitting in DT?

00.

Decision Trees

0. Decision Trees

Introduction

?

What are Decision Trees?

A supervised learning algorithm used for **classification** and **regression**

0. Decision Trees

Introduction

What are Decision Trees?

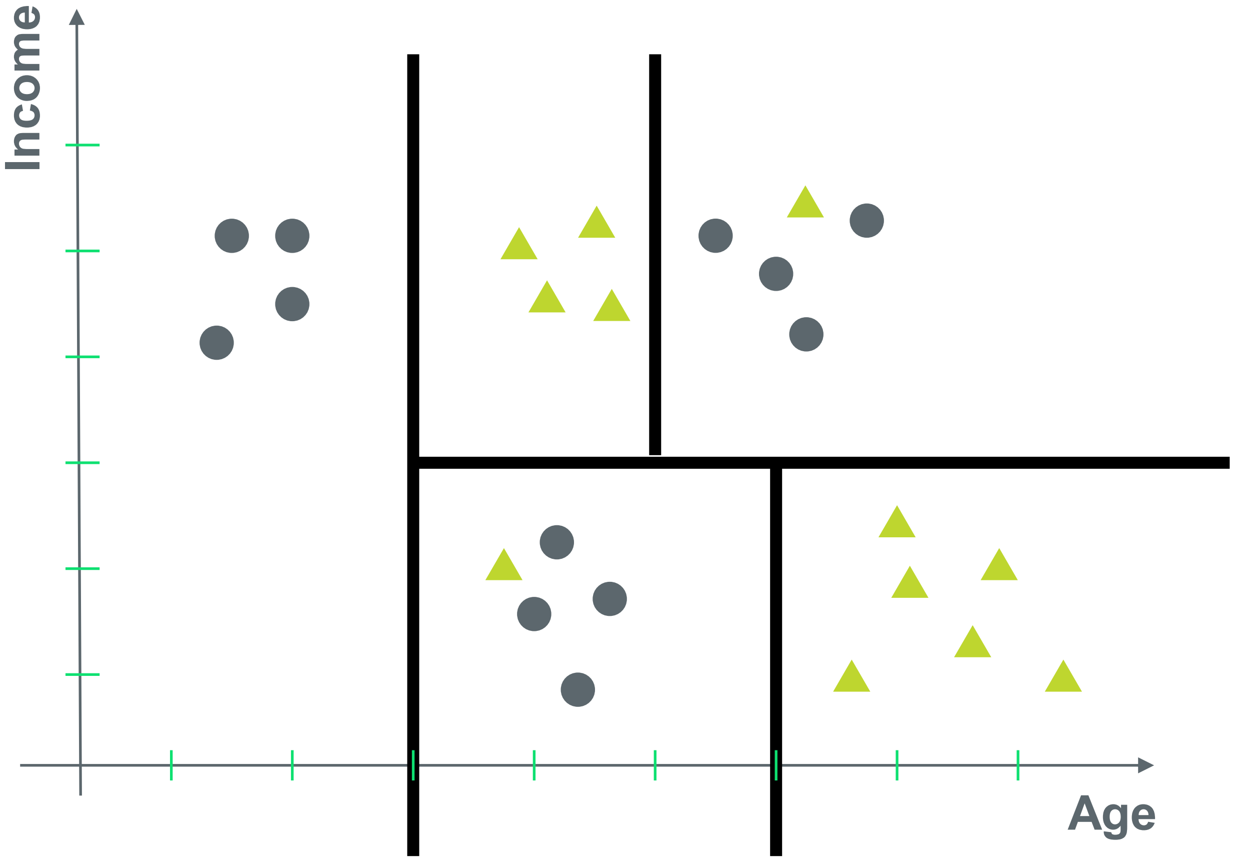
- Decision tree can be though of as **classification** and **regression** tools
- One of its major advantages has to do with the fact that they represent rules, which are fairly simple to interpret

- In some problems we are just interested in achieving the best precision possible. In others we are more interested in understanding the results and the way the model is producing the estimates
  - Sometimes the reasons that underlie certain decisions are of paramount importance.

Interpretability

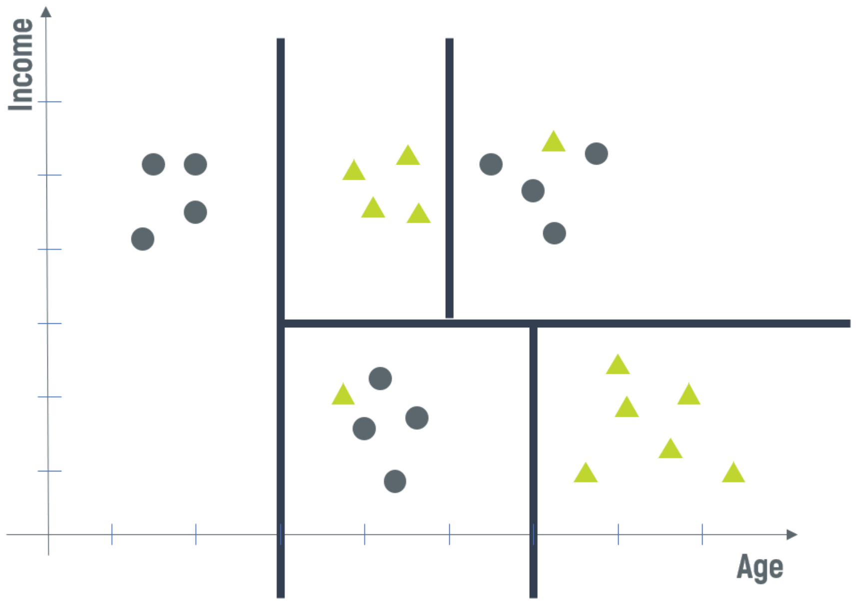
0. Decision Trees

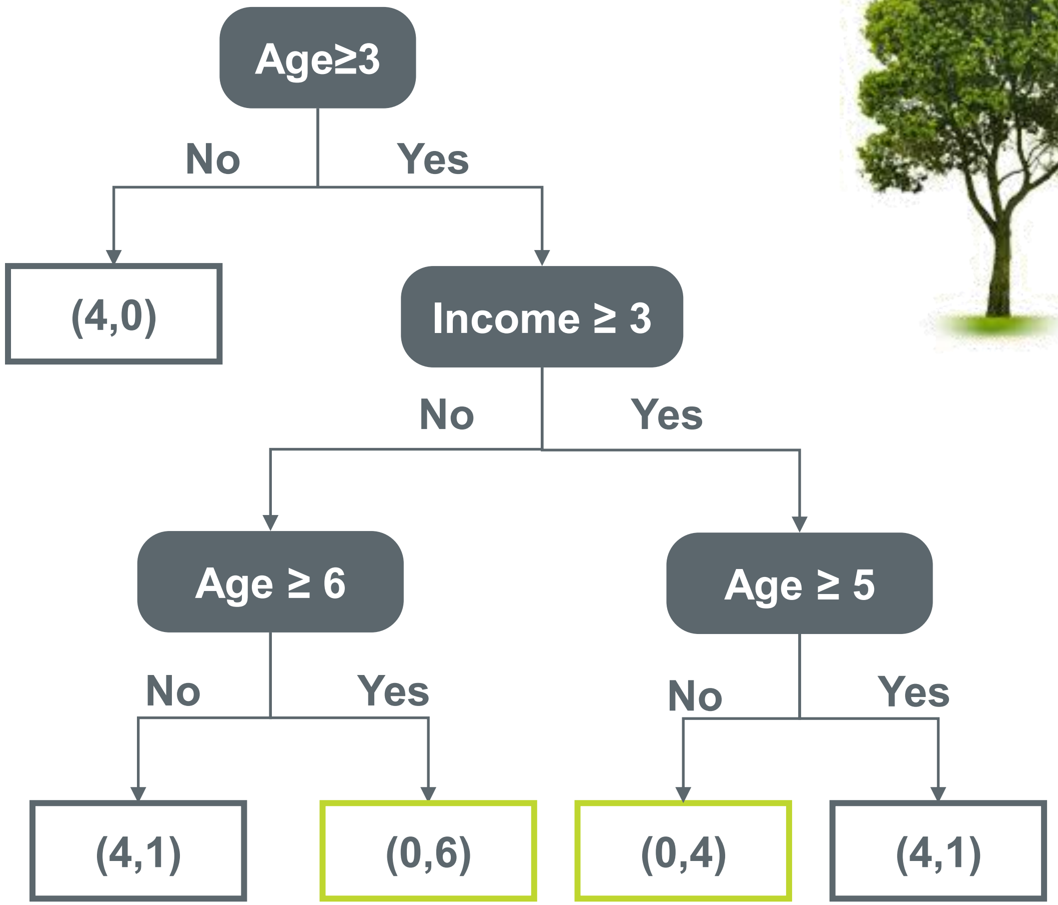
Introduction



0. Decision Trees

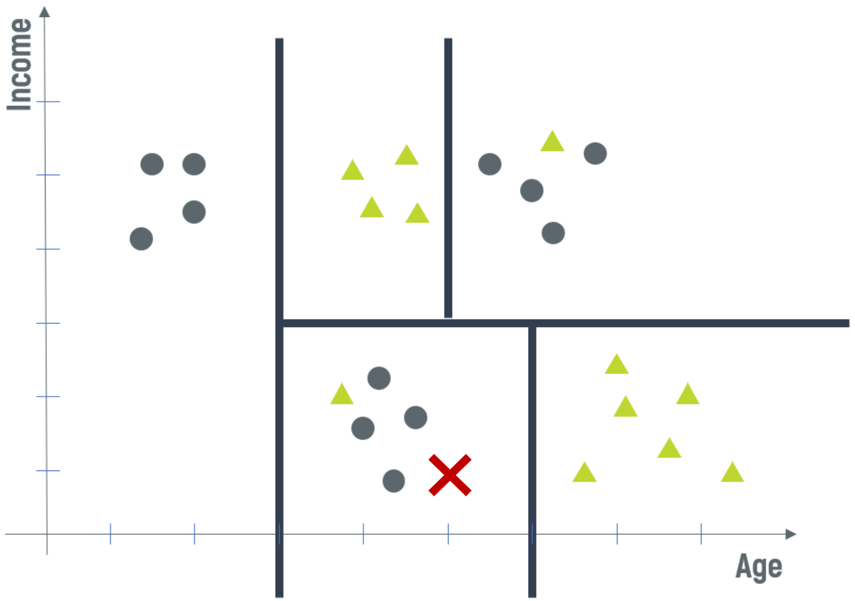
Introduction



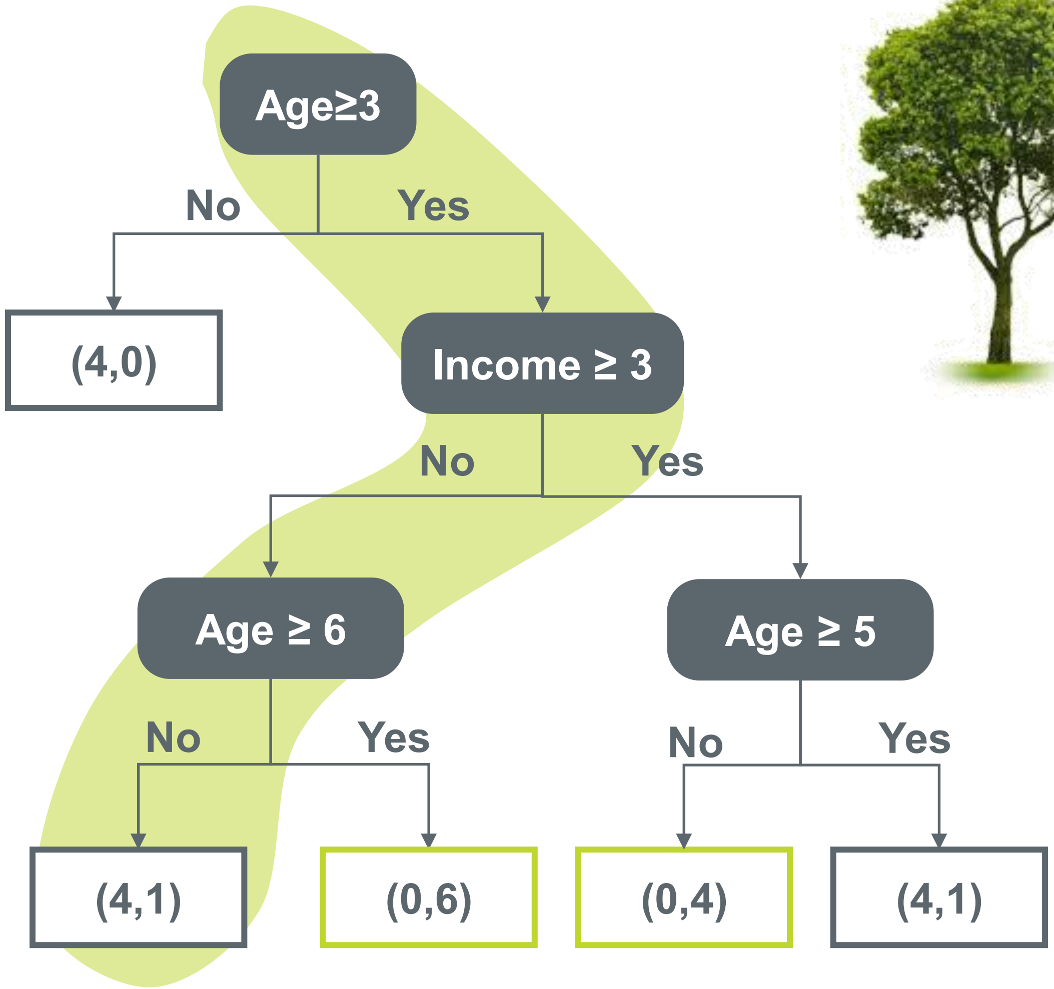


0. Decision Trees

Introduction



What if I have a new observation where age is equal to 5 and income equal to 1?



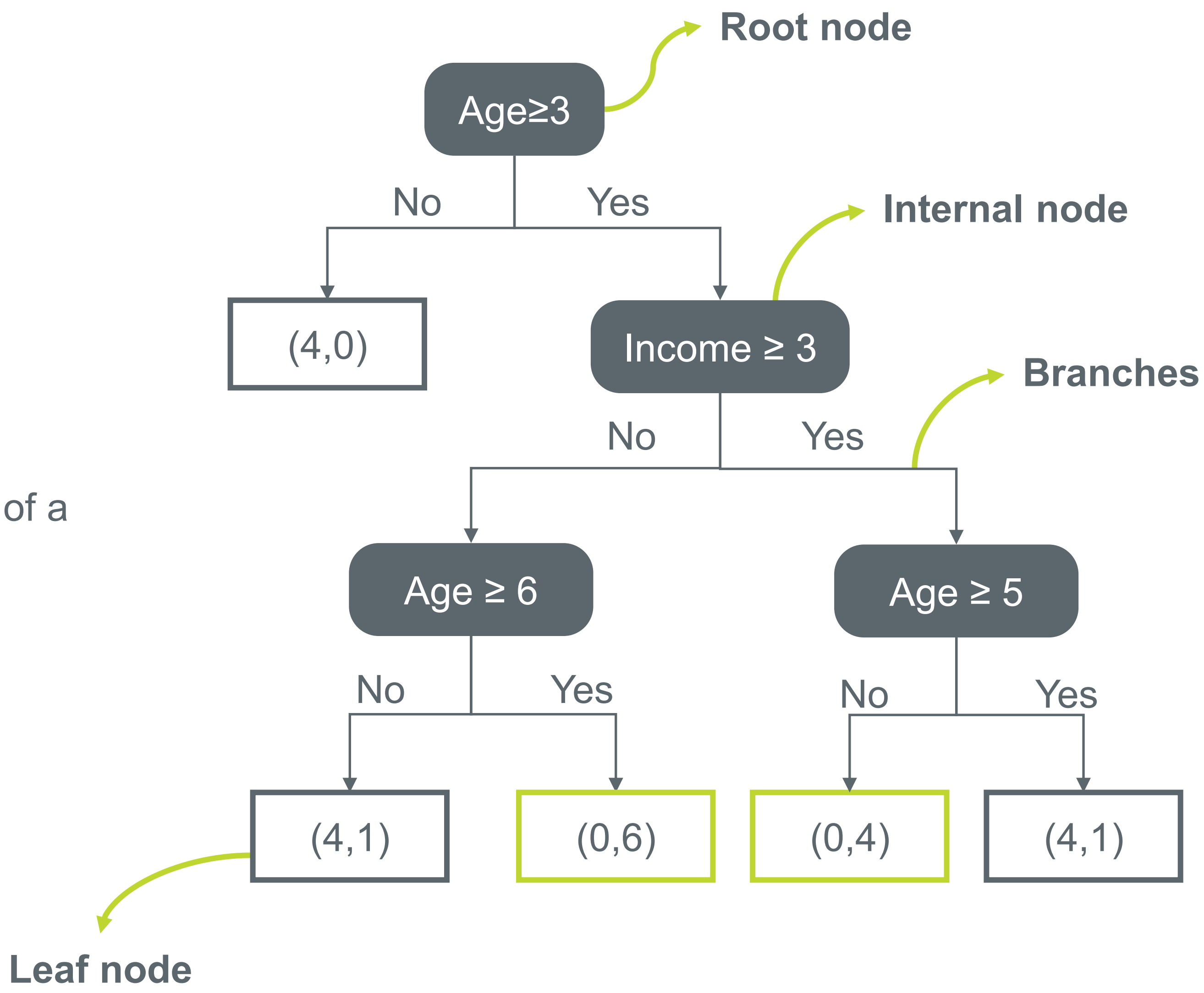
0. Decision Trees

Introduction

The objective in a Classification Tree is to:

- Discriminate between classes
- To obtain leaves that are as pure as possible.

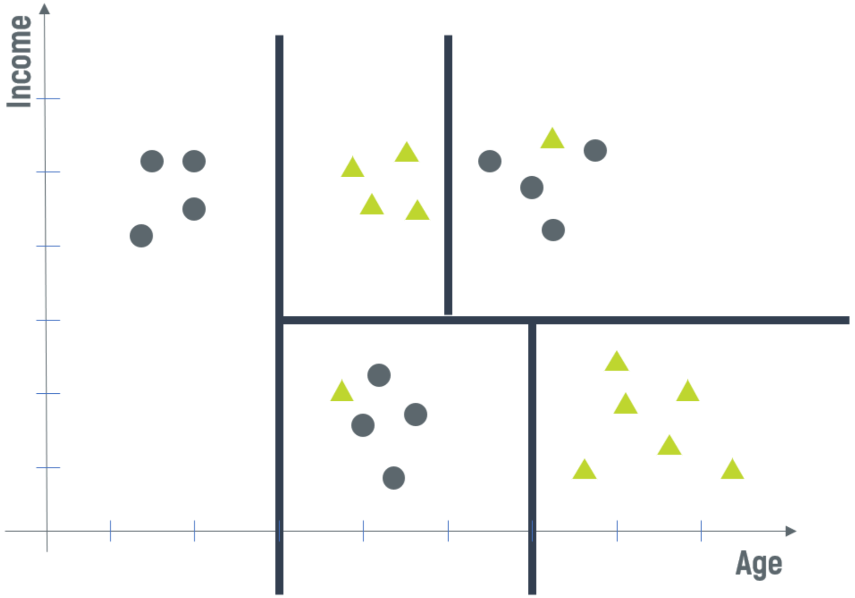
  Hopefully each leaf only represents individuals of a particular class

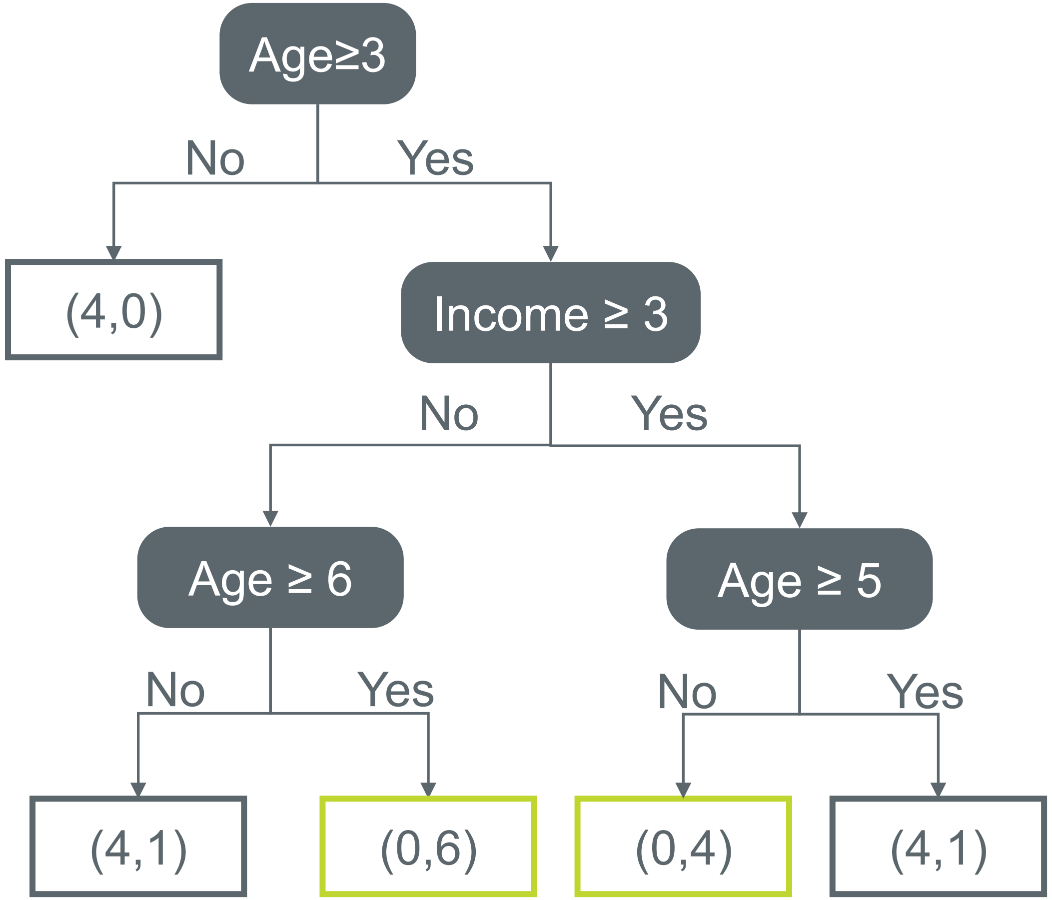


Classification Tree

0.1. Decision Trees

Classification vs Regression Trees

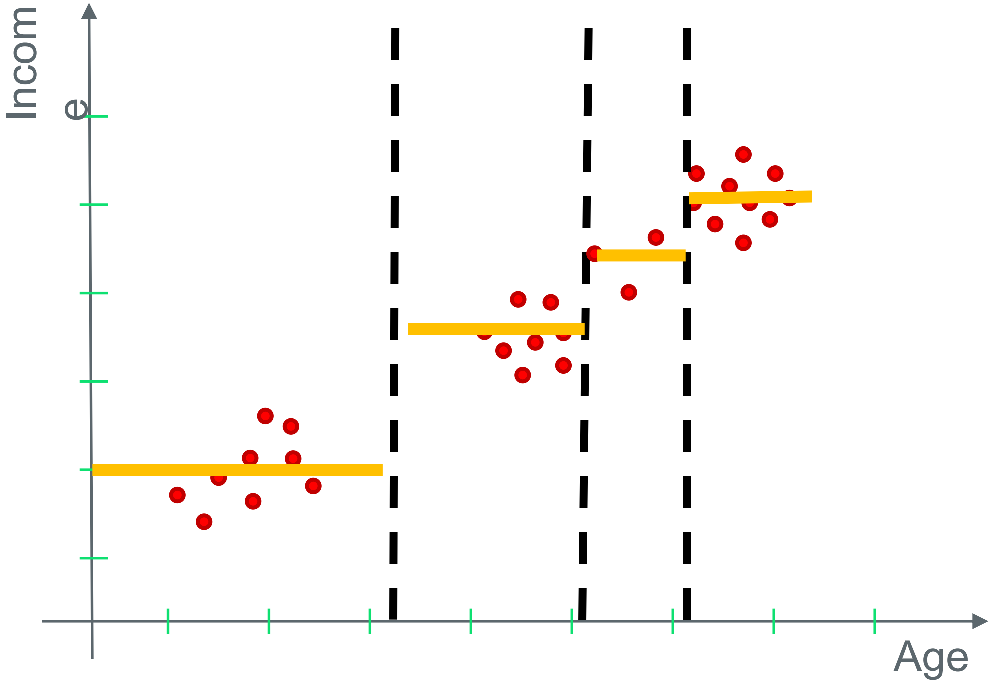


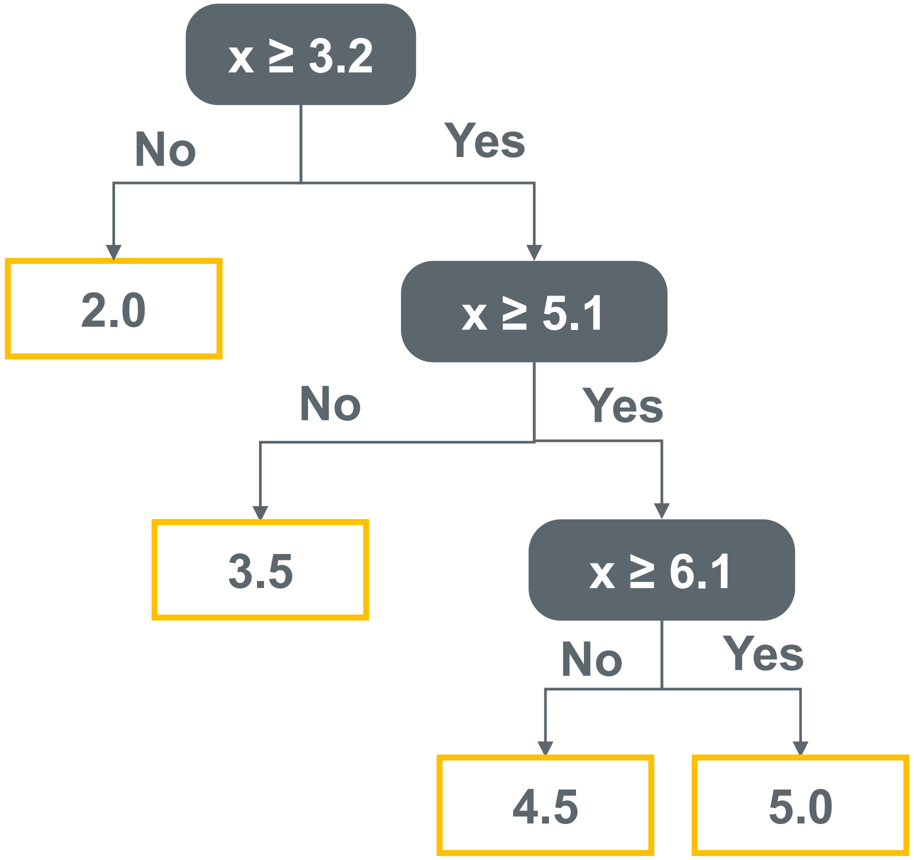


Regression Tree

0. Decision Trees

Introduction



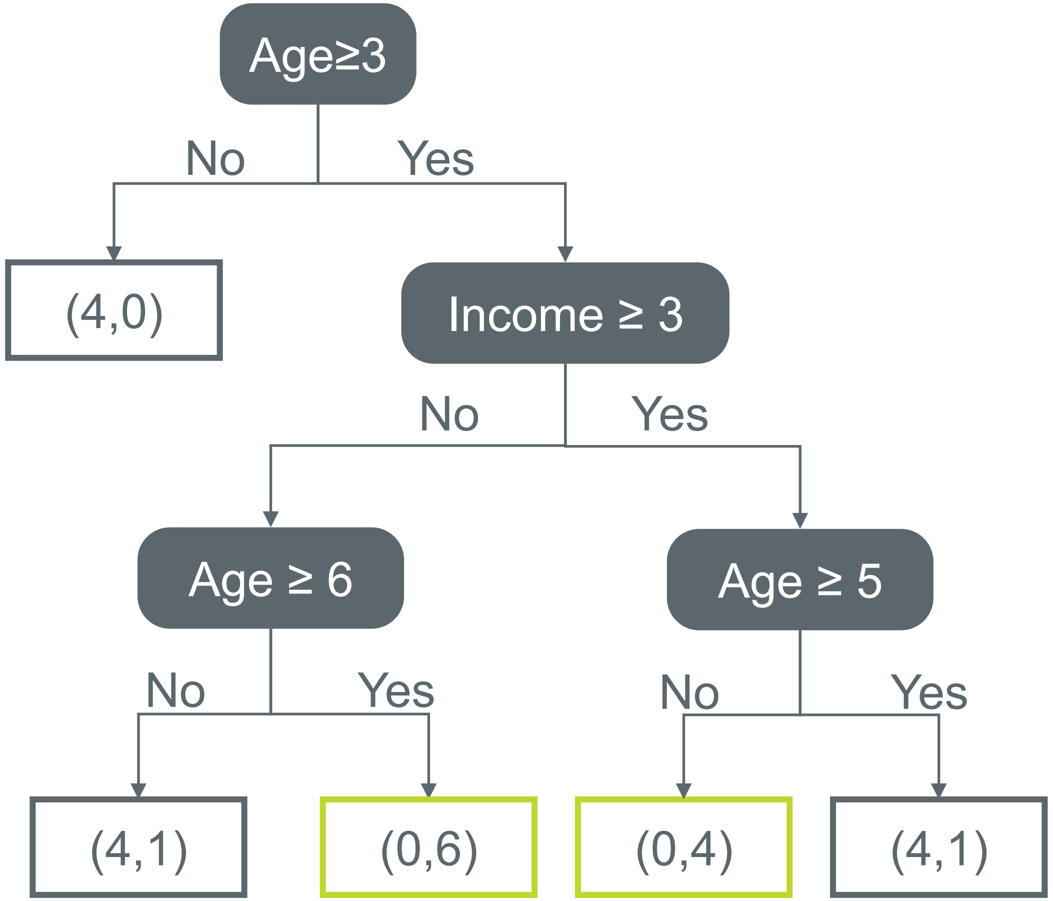


0.2. Decision Trees

Rule Extraction from trees

- Each edge adds a conjunction (∧)
- Each new leaf adds a disjunction (∨)

● $\Leftrightarrow (age < 3)$
● $\Leftrightarrow (age \geq 3) \wedge (income < 3) \wedge (age < 6)$
▲ $\Leftrightarrow (age \geq 3) \wedge (income < 3) \wedge (age \geq 6)$
▲ $\Leftrightarrow (age \geq 3) \wedge (income \geq 3) \wedge (age < 5)$
● $\Leftrightarrow (age \geq 3) \wedge (income \geq 3) \wedge (age \geq 5)$

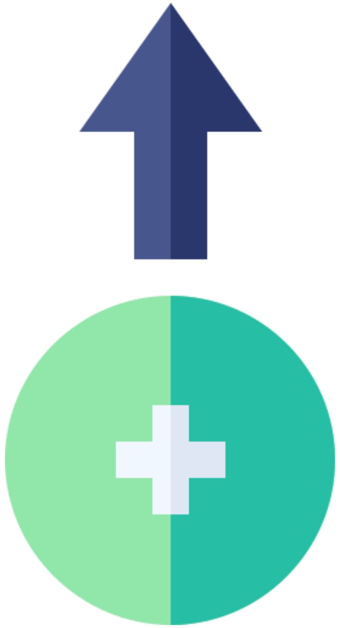


● $\Leftrightarrow (age < 3) \vee \big((age \geq 3) \wedge (income < 3) \wedge (age < 6)\big) \vee ((age \geq 3) \wedge (income \geq 3) \wedge (age \geq 5))$

▲ $\Leftrightarrow \big((age \geq 3) \wedge (income \geq 3) \wedge (age < 5)\big) \vee ((age \geq 3) \wedge (income \geq 3) \wedge (age \geq 5))$

0.4. Decision Trees

Advantages and Disadvantages



Interpretation

- Easily understand the underlying reason to the decision

No problems in dealing with different types of data

- Interval, ordinal, nominal, etc.
- Not necessary to define the relative importance of the variables

Insensitive to scale factors

- Different types of measurements can be used without the need for normalization

Automatic definition of the attributes that are more relevant in each case

- The most relevant attributes appear in the top part of the tree

Can be adapted to regression

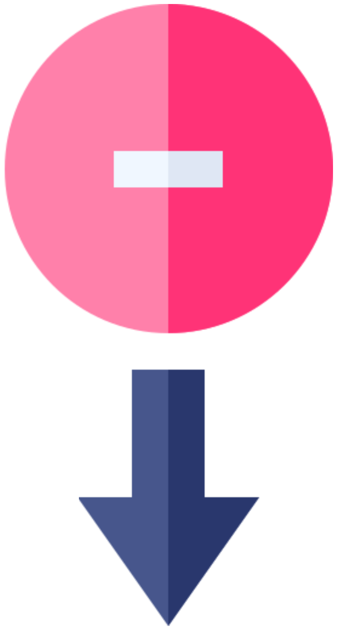
- Linear local models in the leaves

Decision trees are considered a nonparametric method

- No assumptions about the space distribution and the classifier structure

0.4. Decision Trees

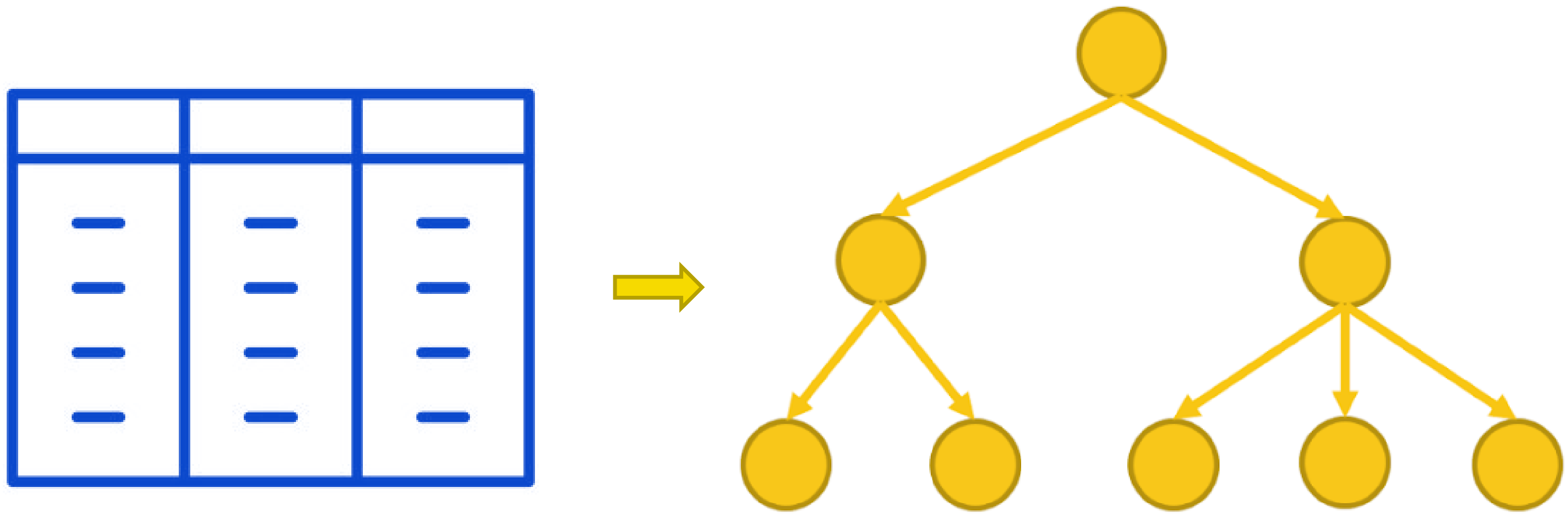
Advantages and Disadvantages



- Most of the algorithms (ID3 and C4.5) require a discrete target
- Small variations in the data can result on very different trees
- Sub-trees can be replicated several times
- Worse results when dealing with many classes
- Linear boundaries perpendicular to the axis

0.5. Decision Trees

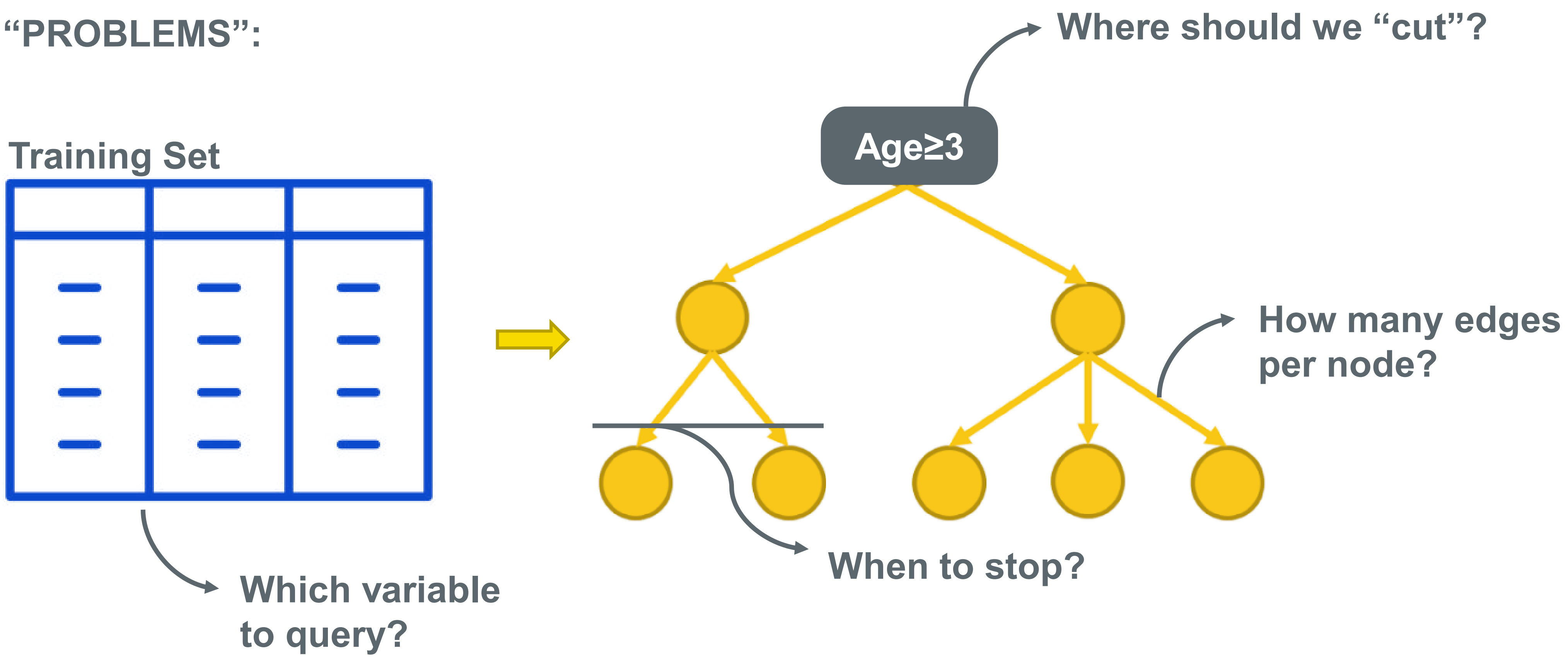
Decision Trees induction – How to build trees?



0.5. Decision Trees

Decision Trees induction – How to build trees?

"PROBLEMS":



0.5. Decision Trees

Decision Trees induction – How to build trees?

BASIC ALGORITHM (a greedy algorithm)

- Tree is constructed in a top-down recursive divide-and-conquer manner
- At start, all the training observations are at the root
- If attributes are continuous-valued, they are discretized in advance (see slide 47)
- Observations are partitioned recursively based on selected attributes

Conditions for stopping partitioning

- All observations for a given node belong to the same class
- There are no remaining attributes for further partitioning – majority voting is employed for classify the leaf
- There are no samples left

0.6. Decision Trees

The algorithms

DECISION TREES

- **DDT**
  Divisive decision tree
  (Hunt 62)
- **ID3, C4.5, C5**
  Iterative Dichotomizer 3
  (Quinlan 86, 93)
- **CART**
  Classification and Regression Trees
  (Breiman 84)
- **CHAID**
  Chi-Squared Automatic Interaction Detection
  (Hartigan 75)
- …

01.

Classification Trees

1. Classification Trees

Introduction

?

What are Classification Trees?

Tree models where the target variable can take a discrete set of values

DDT – Divisive Decision Tree

1.1. Classification Trees

DDT

Major characteristics

- It is a greed search
- There is no “backtracking” (once a partition is done there is no re-evaluation)
- It can become stuck in a local minimum
- It uses discriminate power as selection measure

The algorithm (continue on the next slide)

- Start with a dataset of pre-classified individuals (examples or instances)
- Each node specifies an attribute (independent variables) used as a test
- N – is node N
- ASET – attribute set
- ISET – instance set (individuals or examples)

1.1. Classification Trees

DDT

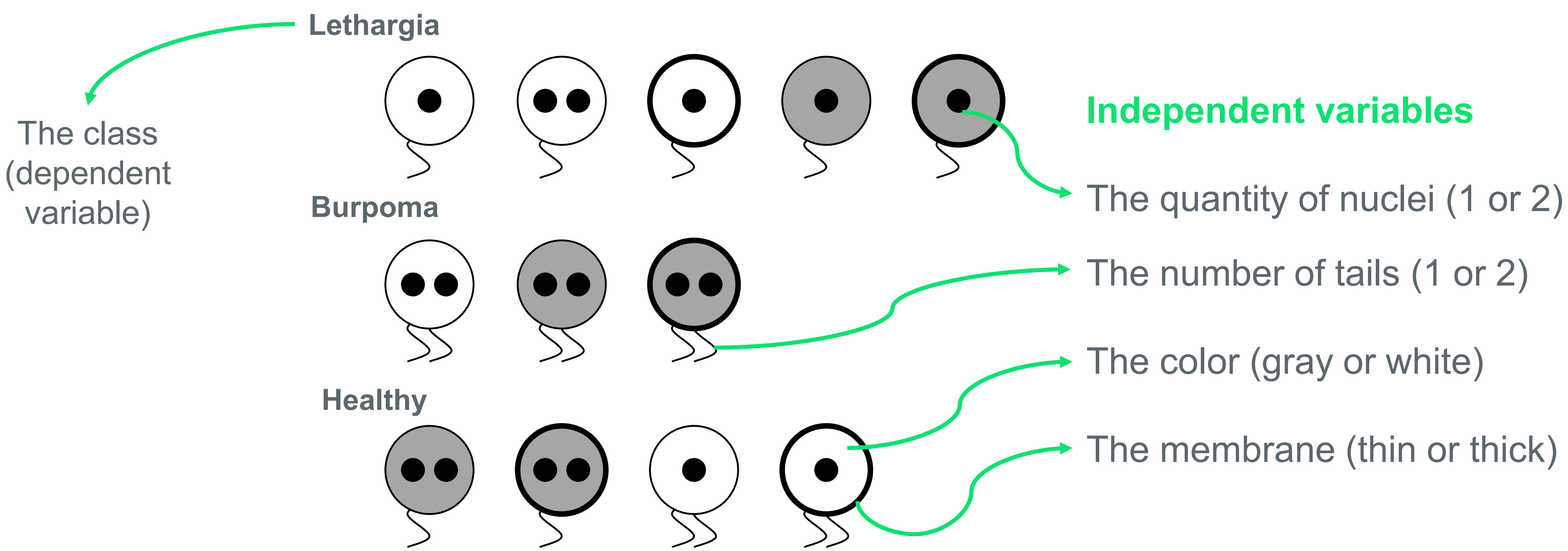
```
DDT(N, ASET, ISET)

If the ISET is empty then
   the terminal node N is an unknown class
elseif all the examples of ISET are of the same class or ASET is empty
 then the terminal node N has the name of the class
else
 For each attribute A of the set of attribute ASET
  Evaluate A according to its capability to discriminate a class
 Select the attribute B which has the best discriminate value
 For each value V of the best attribute B
  Create a new node C from node N
  Place the attribute value pair (B, V) in C
  Let JSET be the set of examples of ISET with value V in B
  Let KSET be the set of attributes of ASET with B removed
  DDT(C, KSET, JSET)
```

Let's see an example…

1.1. Classification Trees

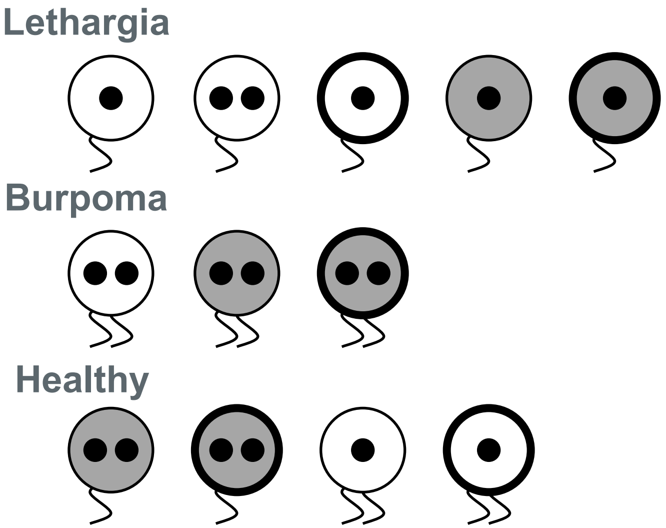
DDT

Example - Cell analysis (Langley 96)



1.1. Classification Trees

DDT



# nuclei	# tails	Color	Membrane	Class
1	1	Light	Thin	**Lethargia**
2	1	Light	Thin	**Lethargia**
1	1	Light	Thick	**Lethargia**
1	1	Dark	Thin	**Lethargia**
1	1	Dark	Thick	**Lethargia**
2	2	Light	Thin	**Burpoma**
2	2	Dark	Thin	**Burpoma**
2	2	Dark	Thick	**Burpoma**
2	1	Dark	Thin	**Healthy**
2	1	Dark	Thick	**Healthy**
1	2	Light	Thin	**Healthy**
1	2	Light	Thick	**Healthy**

1.1. Classification Trees

DDT

The discriminative metric:

Measure the attribute discrimination:

$$Discriminate\ power = \frac{1}{n}\sum_{i=1} C_i$$

-where n is the total number of examples

-C_i is the number of examples correctly classified by the most frequent class

This is a measure of “dominance” or “purity”

1.1. Classification Trees

DDT

# nuclei	# tails	Color	Membrane	Class
1	1	Light	Thin	**Lethargia**
2	1	Light	Thin	**Lethargia**
1	1	Light	Thick	**Lethargia**
1	1	Dark	Thin	**Lethargia**
1	1	Dark	Thick	**Lethargia**
2	2	Light	Thin	**Burpoma**
2	2	Dark	Thin	**Burpoma**
2	2	Dark	Thick	**Burpoma**
2	1	Dark	Thin	**Healthy**
2	1	Dark	Thick	**Healthy**
1	2	Light	Thin	**Healthy**
1	2	Light	Thick	**Healthy**

Using the number of nuclei:

# nuclei	**1**	**2**
Lethargia	**4**	**1**
Burpoma	**0**	**3**
Healthy	**2**	**2**

Discriminate power : $\frac{(4+3)}{12} = 0.58$

1.1. Classification Trees

DDT

# nuclei	# tails	Color	Membrane	Class
1	1	Light	Thin	**Lethargia**
2	1	Light	Thin	**Lethargia**
1	1	Light	Thick	**Lethargia**
1	1	Dark	Thin	**Lethargia**
1	1	Dark	Thick	**Lethargia**
2	2	Light	Thin	**Burpoma**
2	2	Dark	Thin	**Burpoma**
2	2	Dark	Thick	**Burpoma**
2	1	Dark	Thin	**Healthy**
2	1	Dark	Thick	**Healthy**
1	2	Light	Thin	**Healthy**
1	2	Light	Thick	**Healthy**

Using the number of tails:

# tails	1	2
Lethargia	5	0
Burpoma	0	3
Healthy	2	2

Discriminate power : $\frac{(5+3)}{12} = 0.67$

1.1. Classification Trees

DDT

# nuclei	# tails	Color	Membrane	Class
1	1	Light	Thin	Lethargia
2	1	Light	Thin	Lethargia
1	1	Light	Thick	Lethargia
1	1	Dark	Thin	Lethargia
1	1	Dark	Thick	Lethargia
2	2	Light	Thin	Burpoma
2	2	Dark	Thin	Burpoma
2	2	Dark	Thick	Burpoma
2	1	Dark	Thin	Healthy
2	1	Dark	Thick	Healthy
1	2	Light	Thin	Healthy
1	2	Light	Thick	Healthy

Using color:

color	Light	Dark
Lethargia	3	2
Burpoma	1	2
Healthy	2	2

Discriminate power : $\frac{(3+2)}{12} = 0.41$

1.1. Classification Trees

DDT

# nuclei	# tails	Color	Membrane	Class
1	1	Light	Thin	Lethargia
2	1	Light	Thin	Lethargia
1	1	Light	Thick	Lethargia
1	1	Dark	Thin	Lethargia
1	1	Dark	Thick	Lethargia
2	2	Light	Thin	Burpoma
2	2	Dark	Thin	Burpoma
2	2	Dark	Thick	Burpoma
2	1	Dark	Thin	Healthy
2	1	Dark	Thick	Healthy
1	2	Light	Thin	Healthy
1	2	Light	Thick	Healthy

Using membrane:

membrane	Thin	Thick
Lethargia	3	2
Burpoma	2	1
Healthy	2	2

Discriminate power : $\frac{(3+2)}{12} = 0.41$

1.1. Classification Trees

DDT

# nuclei	1	2
Lethargia	4	1
Burpoma	0	3
Healthy	2	2

0.58

# tails	1	2
Lethargia	5	0
Burpoma	0	3
Healthy	2	2

0.67

color	Light	Dark
Lethargia	3	2
Burpoma	1	2
Healthy	2	2

0.41

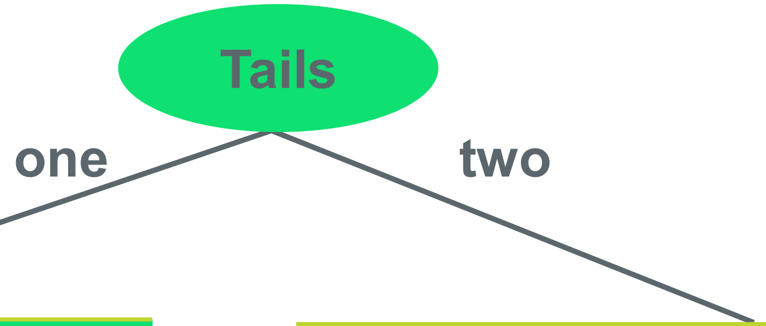
membrane	Thin	Thick
Lethargia	3	2
Burpoma	2	1
Healthy	2	2

0.41

Choice: # tails

1.1. Classification Trees

DDT



# Nuclei	Color	Membrane	Classe
1	Light	Thin	Lethargia
2	Light	Thin	Lethargia
1	Light	Thick	Lethargia
1	Dark	Thin	Lethargia
1	Dark	Thick	Lethargia
2	Dark	Thin	Healthy
2	Dark	Thick	Healthy

# Nuclei	Color	Membrane	Classe
2	Light	Thin	Burpoma
2	Dark	Thin	Burpoma
2	Dark	Thick	Burpoma
1	Light	Thin	Healthy
1	Light	Thick	Healthy

1.1. Classification Trees

DDT

# Nuclei	Color	Membrane	Classe
1	Light	Thin	Lethargia
2	Light	Thin	Lethargia
1	Light	Thick	Lethargia
1	Dark	Thin	Lethargia
1	Dark	Thick	Lethargia
2	Dark	Thin	Healthy
2	Dark	Thick	Healthy

# Nuclei	1	2
Lethargia	4	1
Burpoma	0	0
Healthy	0	2

D.P. = 0.86

Color	Light	Dark
Lethargia	3	2
Burpoma	0	0
Healthy	0	2

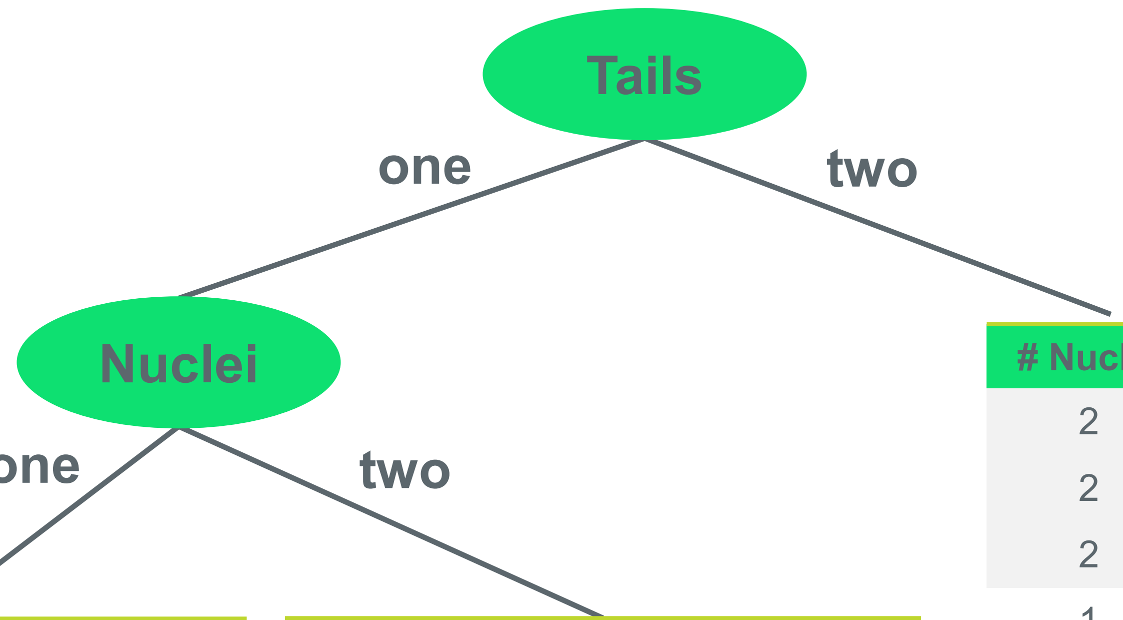
D.P. = 0.71

Membrane	Thin	Thick
Lethargia	3	2
Burpoma	0	0
Healthy	1	1

D.P. = 0.71

1.1. Classification Trees

DDT



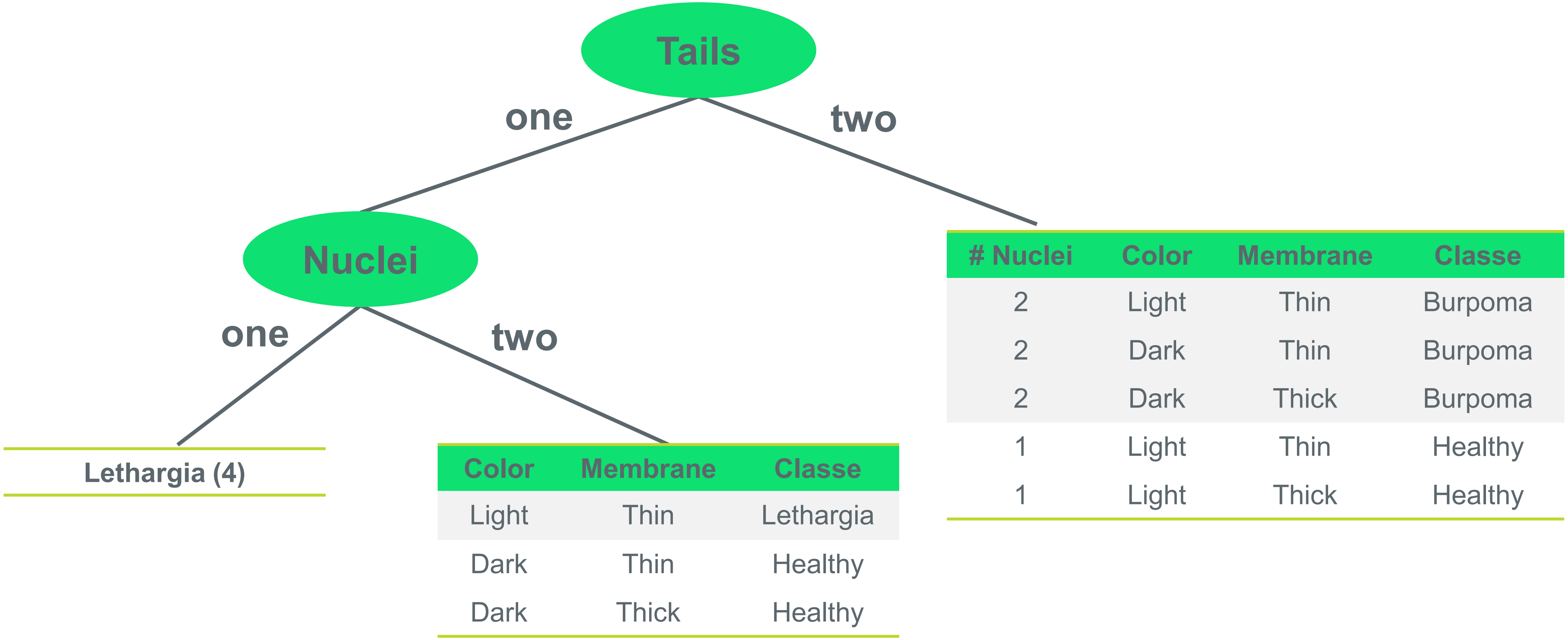
Color	Membrane	Classe
Light	Thin	Lethargia
Light	Thick	Lethargia
Dark	Thin	Lethargia
Dark	Thick	Lethargia

Color	Membrane	Classe
Light	Thin	Lethargia
Dark	Thin	Healthy
Dark	Thick	Healthy

# Nuclei	Color	Membrane	Classe
2	Light	Thin	Burpoma
2	Dark	Thin	Burpoma
2	Dark	Thick	Burpoma
1	Light	Thin	Healthy
1	Light	Thick	Healthy

1.1. Classification Trees

DDT

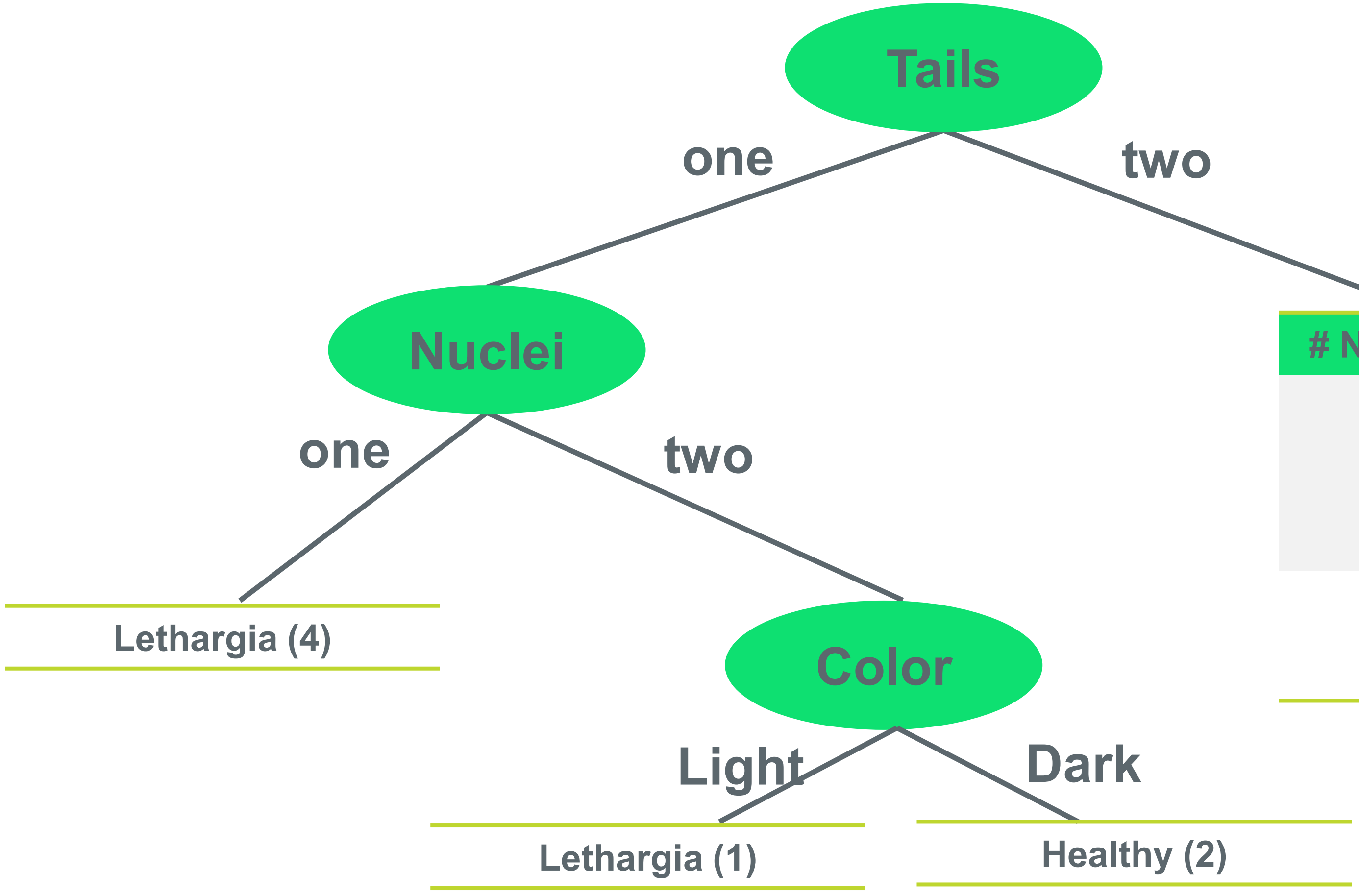


Color	Membrane	Classe
Light	Thin	Lethargia
Dark	Thin	Healthy
Dark	Thick	Healthy

# Nuclei	Color	Membrane	Classe
2	Light	Thin	Burpoma
2	Dark	Thin	Burpoma
2	Dark	Thick	Burpoma
1	Light	Thin	Healthy
1	Light	Thick	Healthy

1.1. Classification Trees

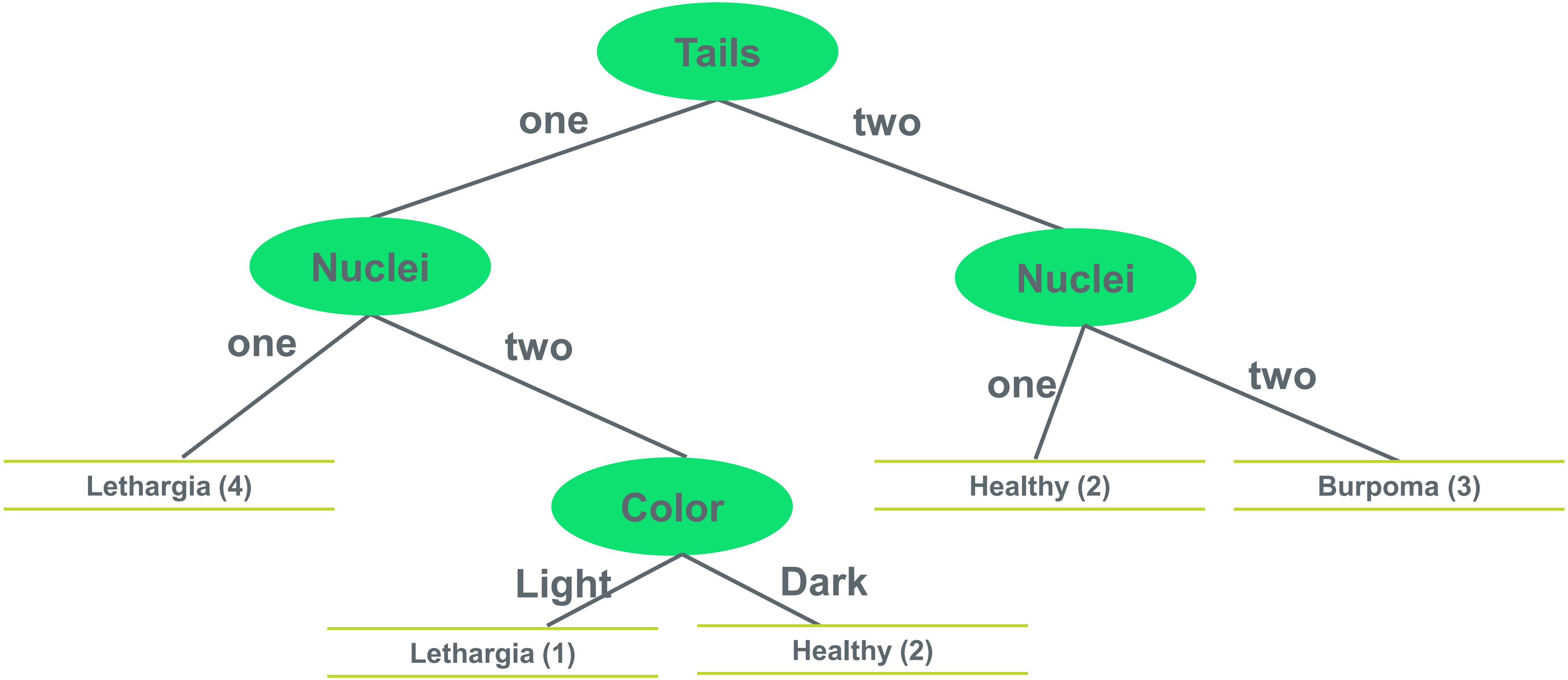
DDT



# Nuclei	Color	Membrane	Classe
2	Light	Thin	Burpoma
2	Dark	Thin	Burpoma
2	Dark	Thick	Burpoma
1	Light	Thin	Healthy
1	Light	Thick	Healthy

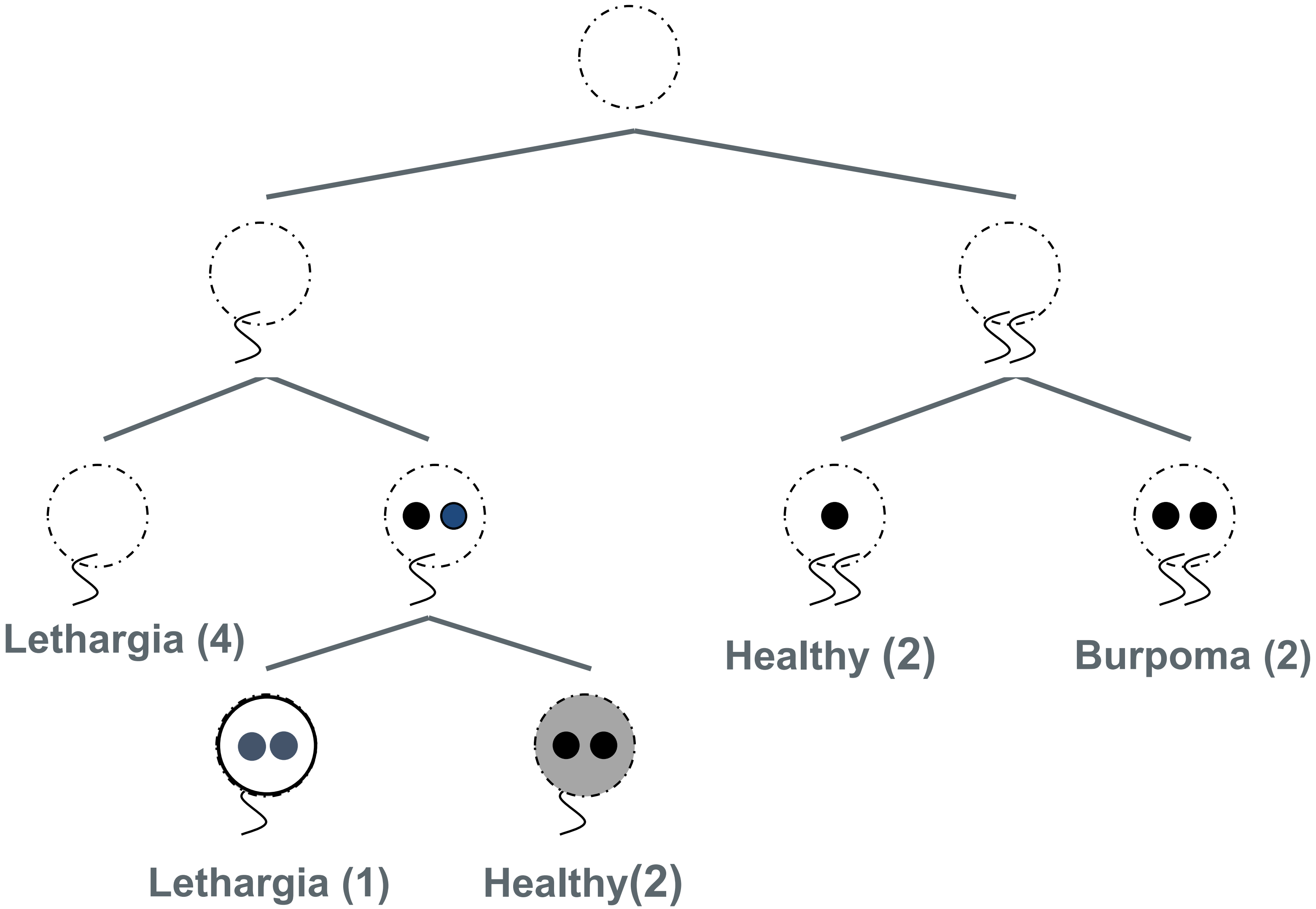
1.1. Classification Trees

DDT



1.1. Classification Trees

DDT



1.1. Classification Trees

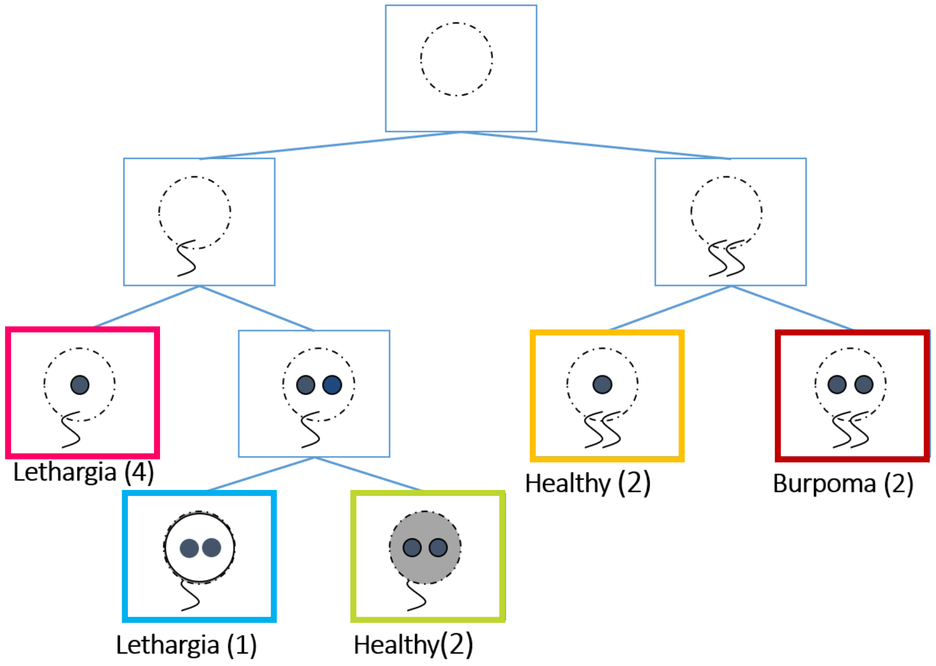
DDT

Rule extraction:

$$(tails = 1) \land (nuclei = 1) \lor (tails = 1) \land (nuclei = 2) \land (color = light) \Rightarrow \boldsymbol{Lethargia}$$

$$(tails = 2) \land (nuclei = 1) \lor (tails = 1) \land (nuclei = 2) \land (color = dark) \Rightarrow \boldsymbol{Healthy}$$

$$(tails = 2) \land (nuclei = 2) \Rightarrow \boldsymbol{Burpoma}$$



ID3 – Iterative Dichotomizer 3

1.2. Classification Trees

ID3

Major characteristics

- It uses **entropy** to measure the "disorder" in each independent variable
- From entropy, we can calculate the **information gain**, the selection measure
- ID3 handles only categorical attributes while C4.5 is able to deal also with numeric values

The nomenclature

Let D be a training set of class-labeled tuples
Let p_i be the probability that an arbitrary tuple in D belongs to class C_i, estimated by $|C_{i,D}|/|D|$
Let $C_{i,D}$ be the set of tuples of class C_i in D.
Let $|C_{i,D}|$ and $|D|$ be the number of tuples in D and $C_{i,D}$ and D

1.2. Classification Trees

ID3

- Expected information (entropy) needed to classify a tuple in D:

$$Entropy(D) = -\sum_{i=1}^{m} p_i \log_2(p_i)$$

- If we choose attribute A to split the node with the tuples in D in v partitions, then the Information needed to classify D is:

$$Entropy_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Entropy(D_j)$$

- The Information gained by branching on attribute A:

$$Gain(A) = Entropy(D) - Entropy_A(D)$$

1.2. Classification Trees

ID3

Example:

age	fever	male	BPM rest	Disease
<=30	high	no	fair	no
<=30	high	no	excellent	no
[31…40]	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
[31…40]	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
[31…40]	medium	no	excellent	yes
[31…40]	high	yes	fair	yes
>40	medium	no	excellent	no

Our dependent variable

- Two classes: Disease = “yes” or “no”

1.2. Classification Trees

ID3

$$Entropy(D) = -\sum_{i=1}^{m} p_i \log_2(p_i)$$

Disease	
Yes	No
9	5

$$Entropy(D) = E(9,5) = -\frac{9}{14}\log_2(\frac{9}{14}) - \frac{5}{14}\log_2(\frac{5}{14}) = 0.940$$

1.2. Classification Trees

ID3

Age	Yes	No	Total	E
<=30	2	3	5	0.971
[31…40]	4	0	4	0
>40	3	2	5	0.971

$Entropy_{Age<30}(D) = E(2{,}3) = -\frac{2}{5}\log_2(\frac{2}{5}) - \frac{3}{5}\log_2(\frac{3}{5})$ = 0.971

$Entropy_{Age[31..40]}(D) = E(4{,}0) = -\frac{4}{4}\log_2(\frac{4}{4}) - \frac{0}{4}\log_2(\frac{0}{4}) = 0$

$Entropy_{Age>40}(D) = E(3{,}2) = -\frac{3}{5}\log_2(\frac{3}{5}) - \frac{2}{5}\log_2(\frac{2}{5}) = 0.971$

$$Entropy_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Entropy(D_j)$$

$$Entropy_{age}(D) = \frac{5}{14}Entropy(2{,}3) + \frac{4}{14}Entropy(4{,}0) + \frac{5}{14}E(3{,}2) = 0.694$$

Note: $\frac{5}{14}Entropy(2{,}3)$ means “age <=30” has 5 out of 14 samples, with 2 yes and 3 no

1.2. Classification Trees

ID3

$$Gain(A) = Entropy(D) - Entropy_A(D)$$

$$Gain(age) = Entropy(D) - Entropy_{age}(D) = 0.940 - 0.694 = 0.246$$

$$Gain(fever) = 0.029$$

$$Gain(male) = 0.151$$

$$Gain(BPM\ rest) = 0.048$$

Since **age** has the highest information gain it is the selected attribute

1.2. Classification Trees

ID3

Age

age	fever	male	BPM rest	Disease
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes

age	fever	male	BPM rest	Disease
[31…40]	high	no	fair	yes
[31…40]	low	yes	excellent	yes
[31…40]	medium	no	excellent	yes
[31…40]	high	yes	fair	yes
[31…40]	high	no	fair	yes

age	fever	male	BPM rest	Disease
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

1.3. Classification Trees

C4.5

C4.5

C4.5

But what about if I want to use the original age (algorithm C4.5)?

Originally Age is a continuous-valued attribute

Must determine the best split point for Age

- Sort the values of Age
- Every midpoint between each pair of adjacent values is a possible split point
- Evaluate $Entropy_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Entropy(D_j)$ with two partitions (Partition 1<split point<Partition 2)
- The point with the minimum entropy for Age is selected as the split-point

age
18
20
22
26
27
29
31
35
41
42
45
48
49
…

21

1.4. Classification Trees

CART

CART

1.4. Classification Trees

CART

-If a data set D contains examples from n classes, **Gini index is defined as:**

$$gini(D) = 1 - \sum_{j=1}^{n} p_j^2$$

where p_j is the relative frequency of class j in D

- Similar to entropy but without the log $\longrightarrow$ **FASTER !**

1.4. Classification Trees

CART

-If a data set D is split on Age into two subsets D1 and D2, the gini index is defined as:

$$gini_{Age}(D) = \frac{|D_1|}{|D|}\, gini\,(D_1) + \frac{|D_2|}{|D|}\, gini\,(D_2)$$

-The reduction in Impurity is given by:

$$\Delta gini(A) = gini(D) \;-\; gini_A(D)$$

The attribute that maximizes the reduction in impurity is selected as splitting attribute

1.4. Classification Trees

CART

$$gini(D) = 1 - \sum_{j=1}^{n} p_j^2$$

For our example:

Disease	
Yes	No
9	5

$$gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

For the variable Age:

Age	Yes	No	Total	Gini
<=30	2	3	5	0.48
[31…40]	4	0	4	0
>40	3	2	5	0.48

$$Gini(D_{age<30}) = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.48$$

…

…

1.4. Classification Trees

CART

$$gini_{Age}(D) = \frac{|D_1|}{|D|} gini\ (D_1) + \frac{|D_2|}{|D|} gini\ (D_2)$$

Age	Yes	No	Total	Gini
<=30	2	3	5	0.48
[31…40]	4	0	4	0
>40	3	2	5	0.48

$$Gini_{age}(D) = \frac{5}{14} Gini(2,3) + \frac{4}{14} Gini(4,0) + \frac{5}{14} Gini(3,2) = 0.343$$

$$\Delta gini(A) = gini(D)\ -\ gini_A(D)$$

$$\Delta gini(Age) = gini(D) - gini_{Age}(D) = 0.459 - 0.343 = 0.116$$

02.

Regression Trees

2. Regression Trees

Introduction

?

What are Regression Trees?

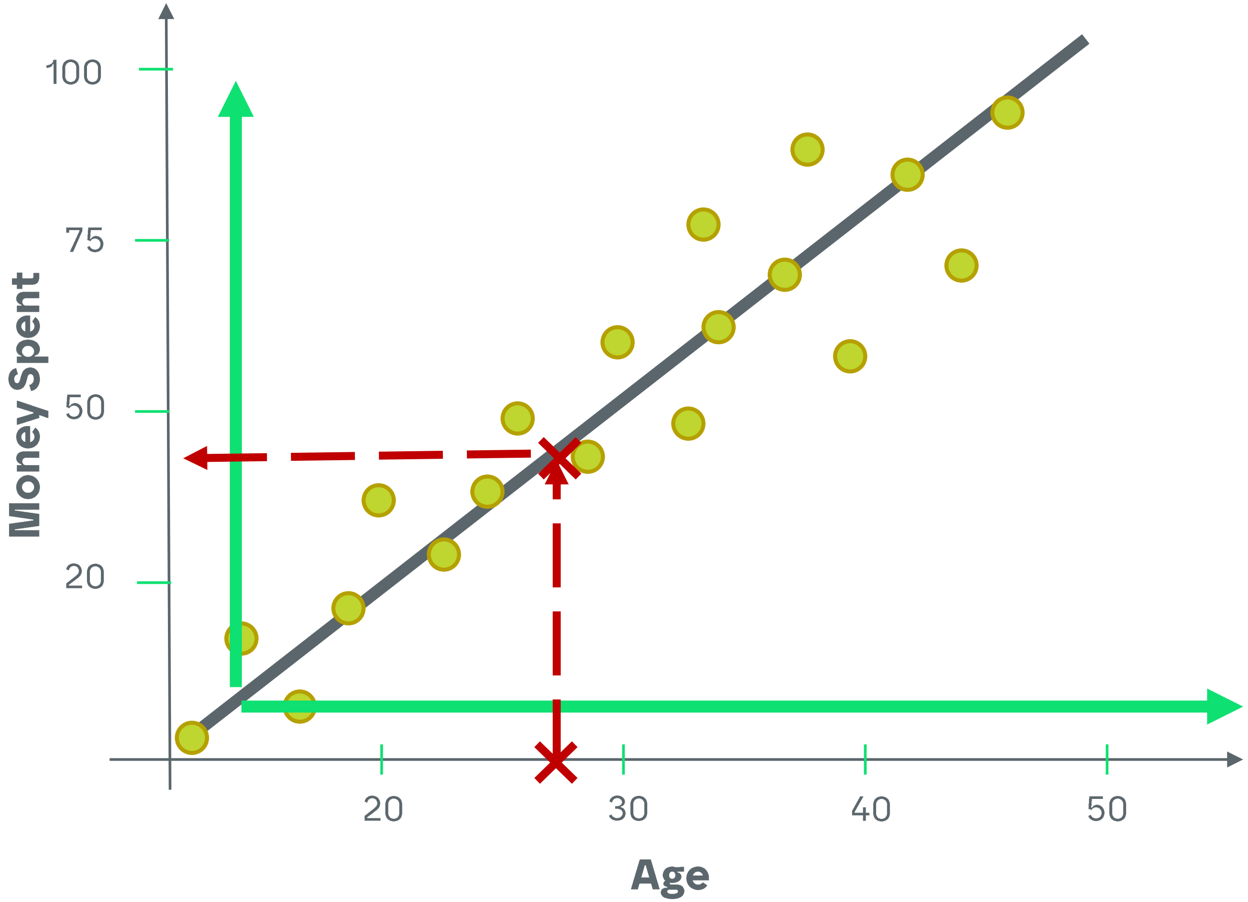
Tree models where the target variable can take a continuous set of values

2. Regression Trees

Introduction

In a regression problem...

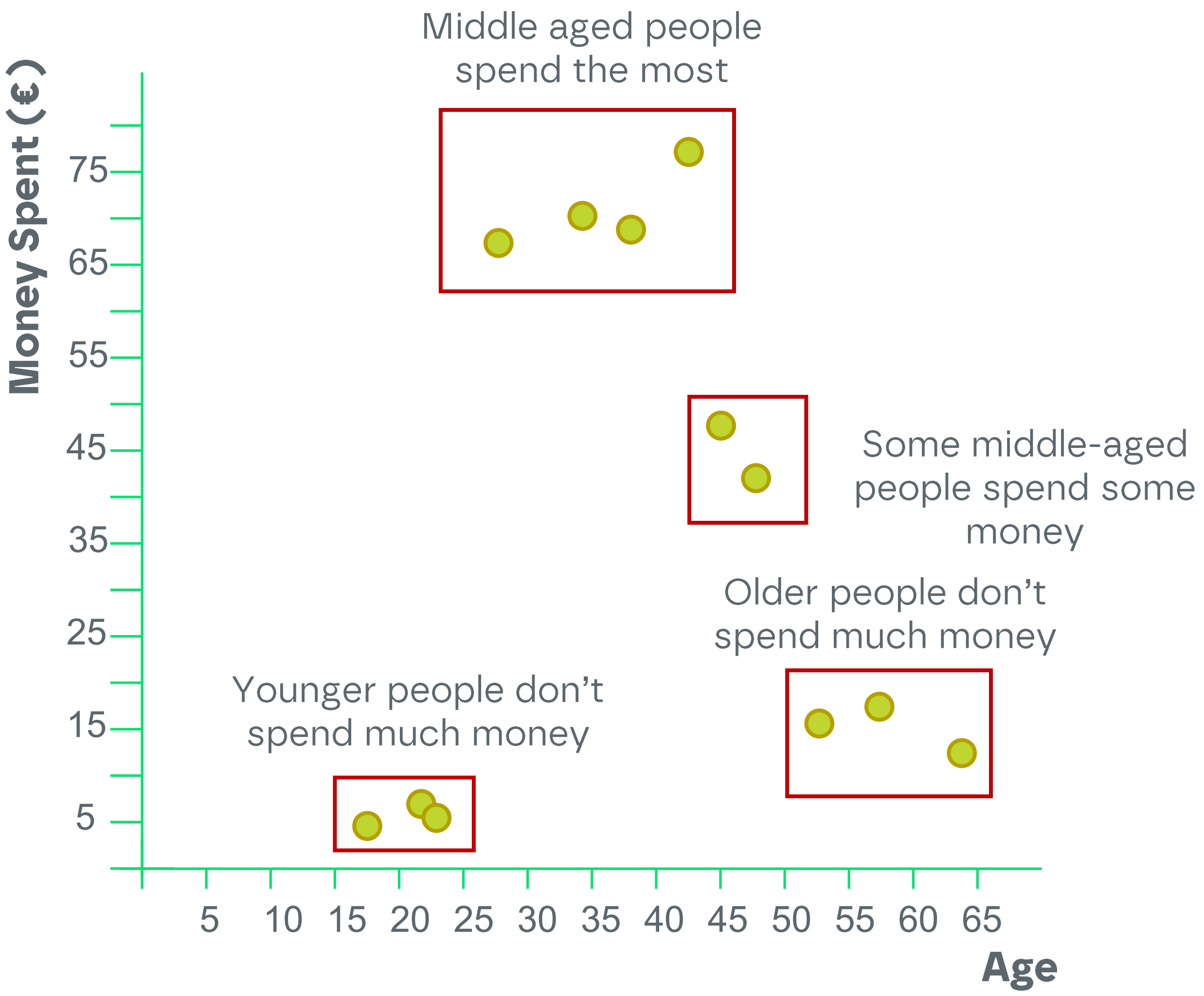
There are some problems that are easily fitted with a linear regression...



2. Regression Trees

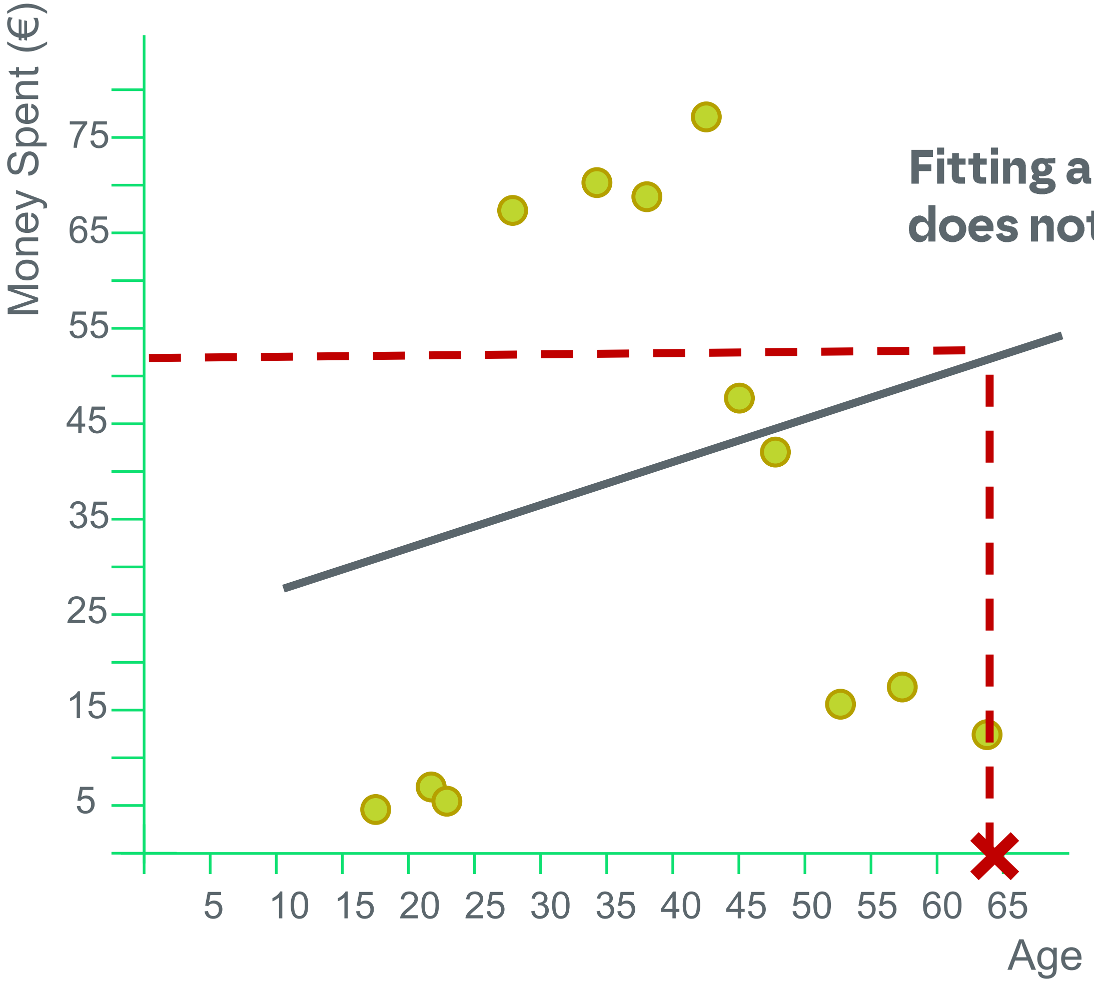
Introduction

Age	Money Spent
18	5
22	7
23	6
27	68
34	72
38	69
43	77
45	48
48	42
53	16
57	18
64	14



2. Regression Trees
Introduction

A problem with one variable...



Fitting a straight line to the data does not seem very useful!

As an example, if someone has the age of 64...

we will predict that the money spent would be around 50€...

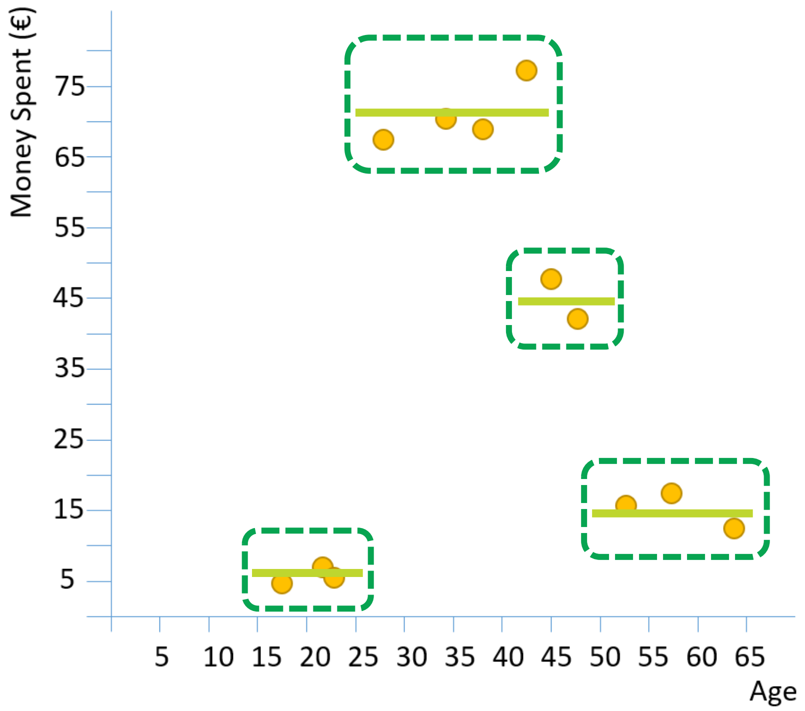
The observed data however suggests it should be around 15€...

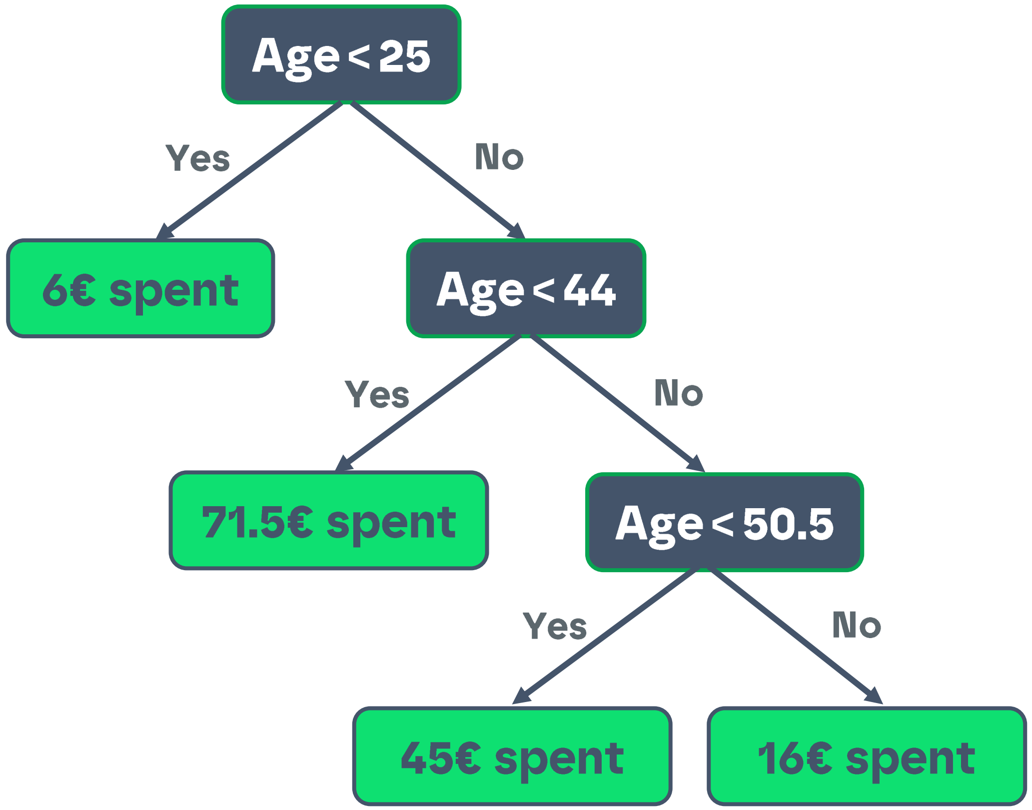
Vary far from reality!

2. Regression Trees

Introduction

Regression tree with one variable…





However, I could obtain the same answer just by looking to the plot!

2. Regression Trees

Introduction

Regression tree with more variables…

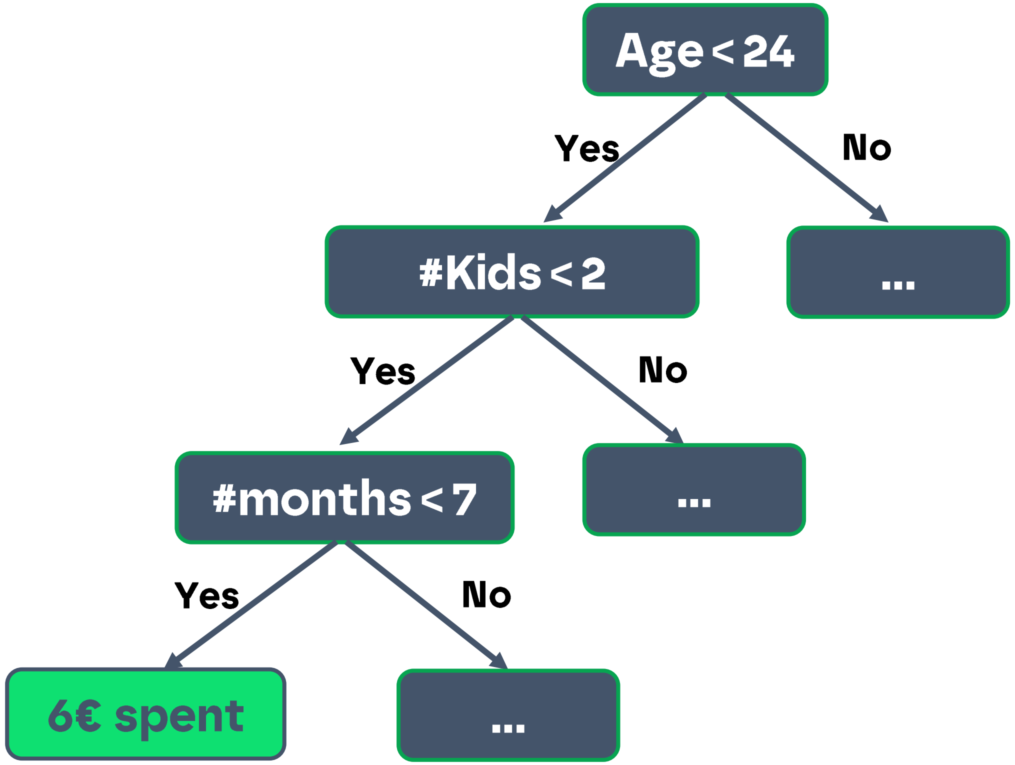
When we have more than 2 predictors, such as Age, Number of Kids and Number of months as client, to predict the money spent through drawing a plot is very difficult, if not impossible

Age	#Kids	#Months as customer	…	Money Spent
18	0	1	…	5
22	1	3	…	7
23	0	6	…	6
27	2	22	…	68
34	3	27	…	72
38	2	29	…	69
43	1	15	…	77
45	0	12	…	48
48	3	17	…	42
53	2	18	…	16
57	3	43	…	18
64	1	32	…	14

2. Regression Trees
Introduction

Regression tree with more variables…
A regression tree easily supports more than 2 predictors

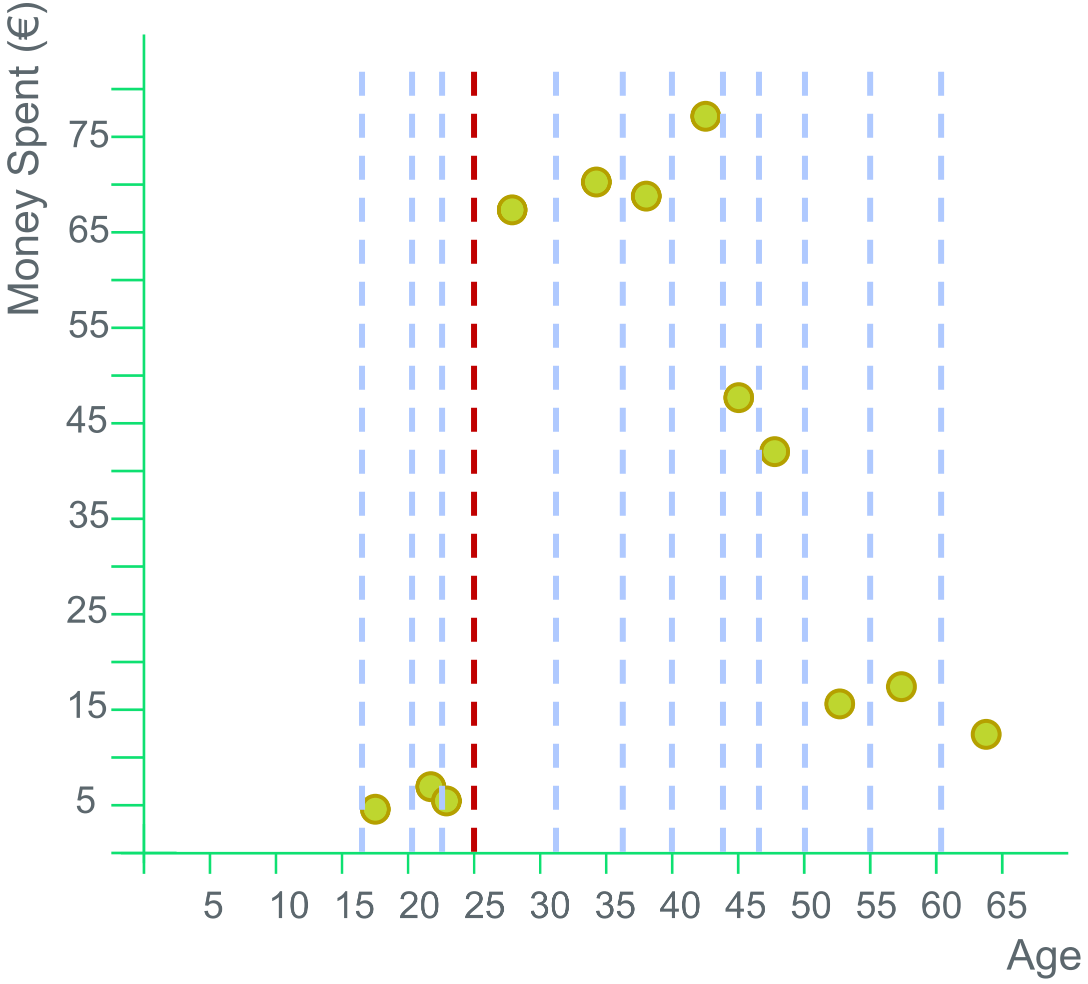
Age	#Kids	#Months as customer	…	Money Spent
18	0	1	…	5



2. Regression Trees

Introduction

Choose the best split



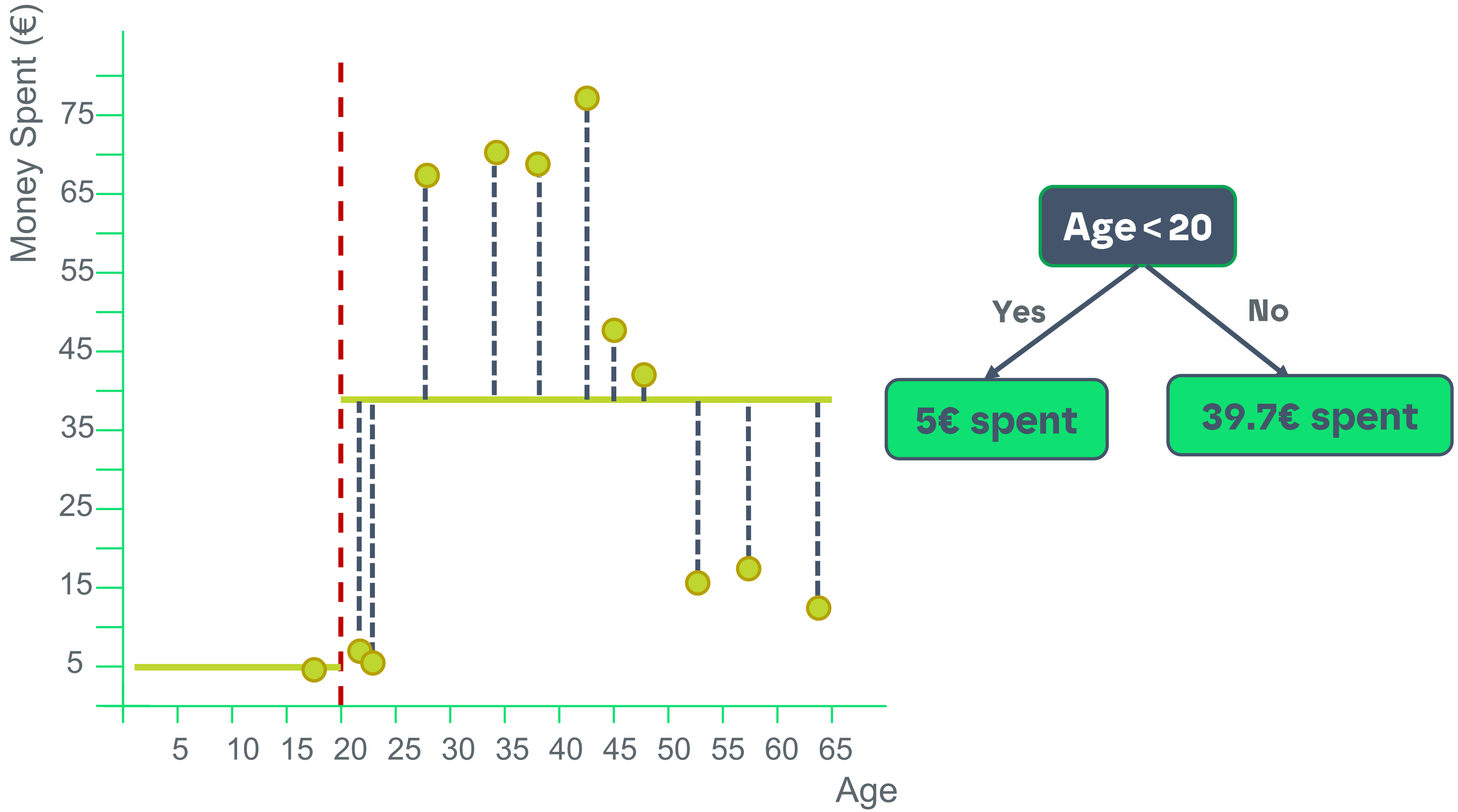
Why do we start with age < 25?

What is the best splitting point?

2. Regression Trees

Introduction

Choose the best split

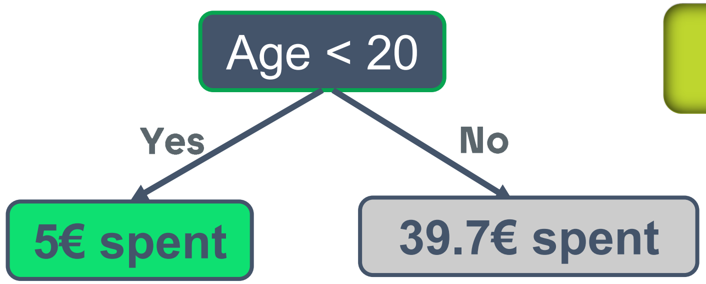


2.1. Regression Trees

Using MSE

MSE

Choose the best split using MSE



$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y})^2$$

Age	Money Spent (y)
18	5
22	7
23	6
...	...
64	14

$$\bar{y} = 5$$

$$MSE\ (Age < 20) = (5 - 5)^2 = 0$$

$$\bar{y} = \frac{7 + 6 + \ ... + 14}{11} = 39.7$$

$$MSE\ (Age \geq 20) = \frac{1}{11} \times ((7 - 39.7)^2 + (6 - 39.7)^2 + \ ... \ + (14 - 39.7)^2) = 733.3$$

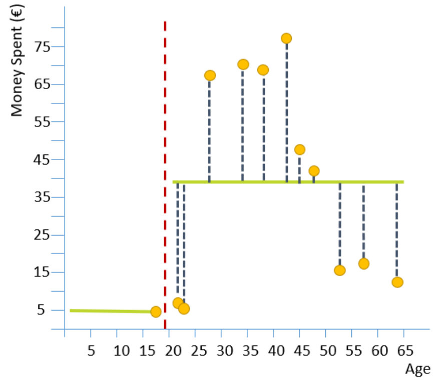
$$TOTAL\ MSE = 0 + 733.3 = 733.3$$

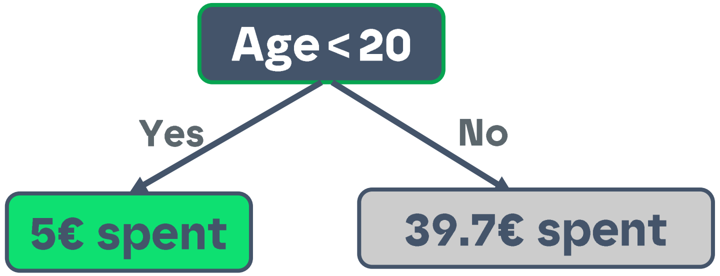
MSE

2.1. Regression Trees

Using MSE

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y})^2$$

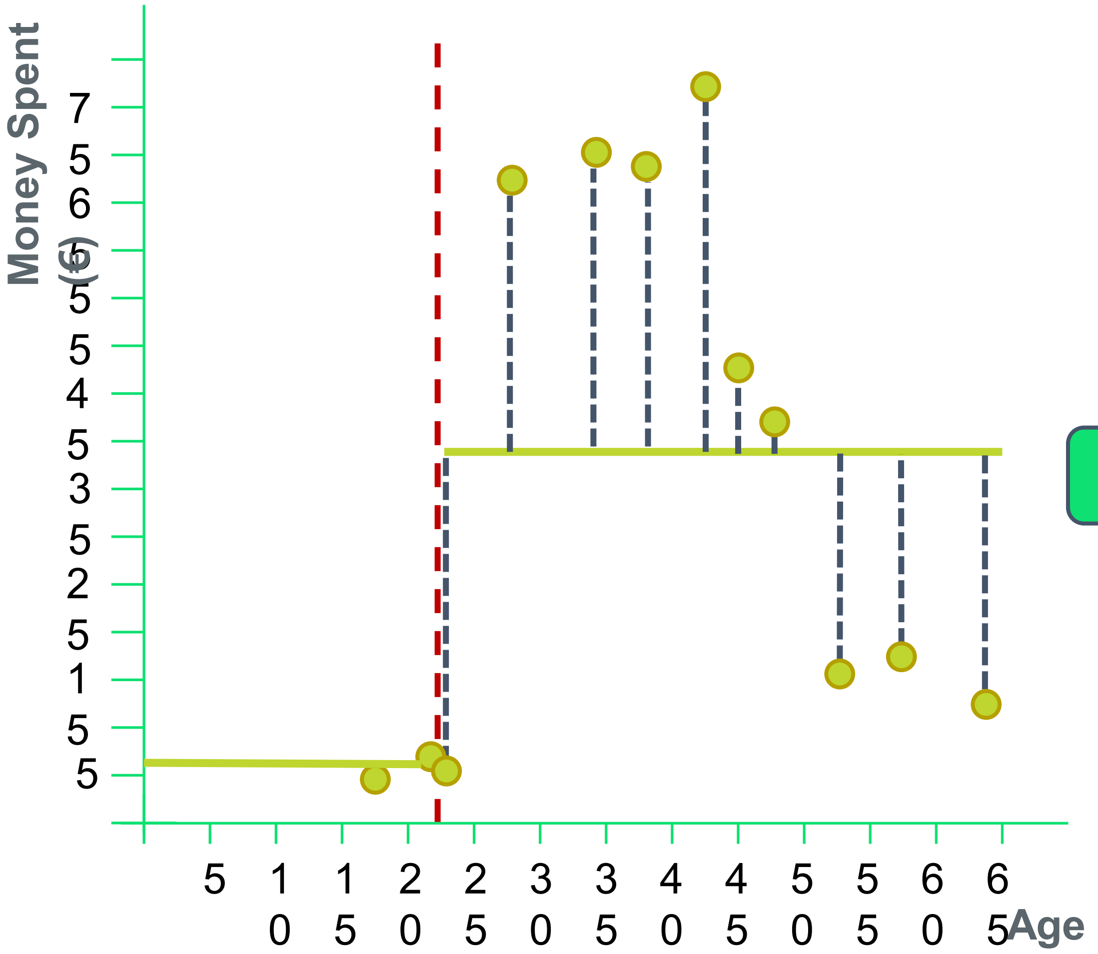


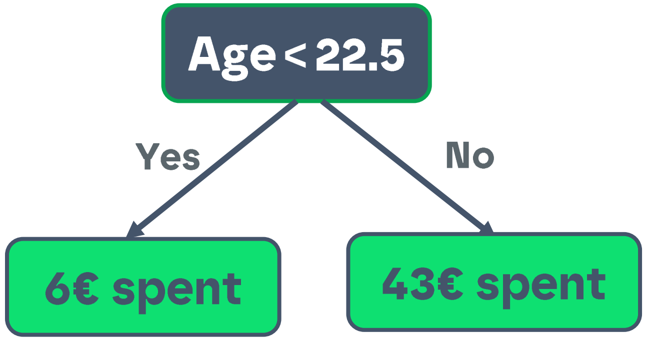


Age	Money Spent (y)	$\hat{y}$	MSE
18	5	5	0.0
22	7	39.7	733.3
23	6	39.7	
27	68	39.7	
34	72	39.7	
38	69	39.7	
43	77	39.7	
45	48	39.7	
48	42	39.7	
53	16	39.7	
57	18	39.7	
64	14	39.7	
		TOTAL MSE	733.3

2.1. Regression Trees

Using MSE



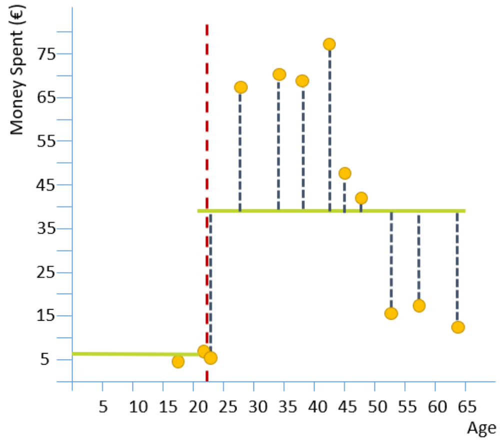


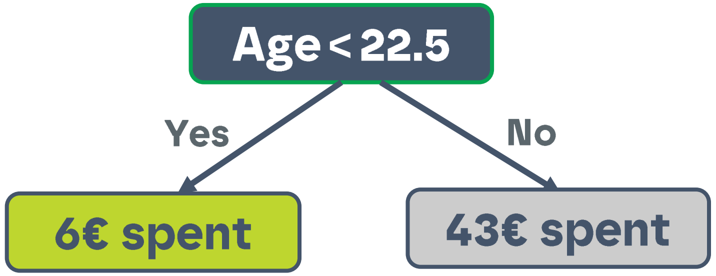
MSE

2.1. Regression Trees

Using MSE

$$MSE = \frac{1}{n}(y_i - \hat{y}_i)^2$$



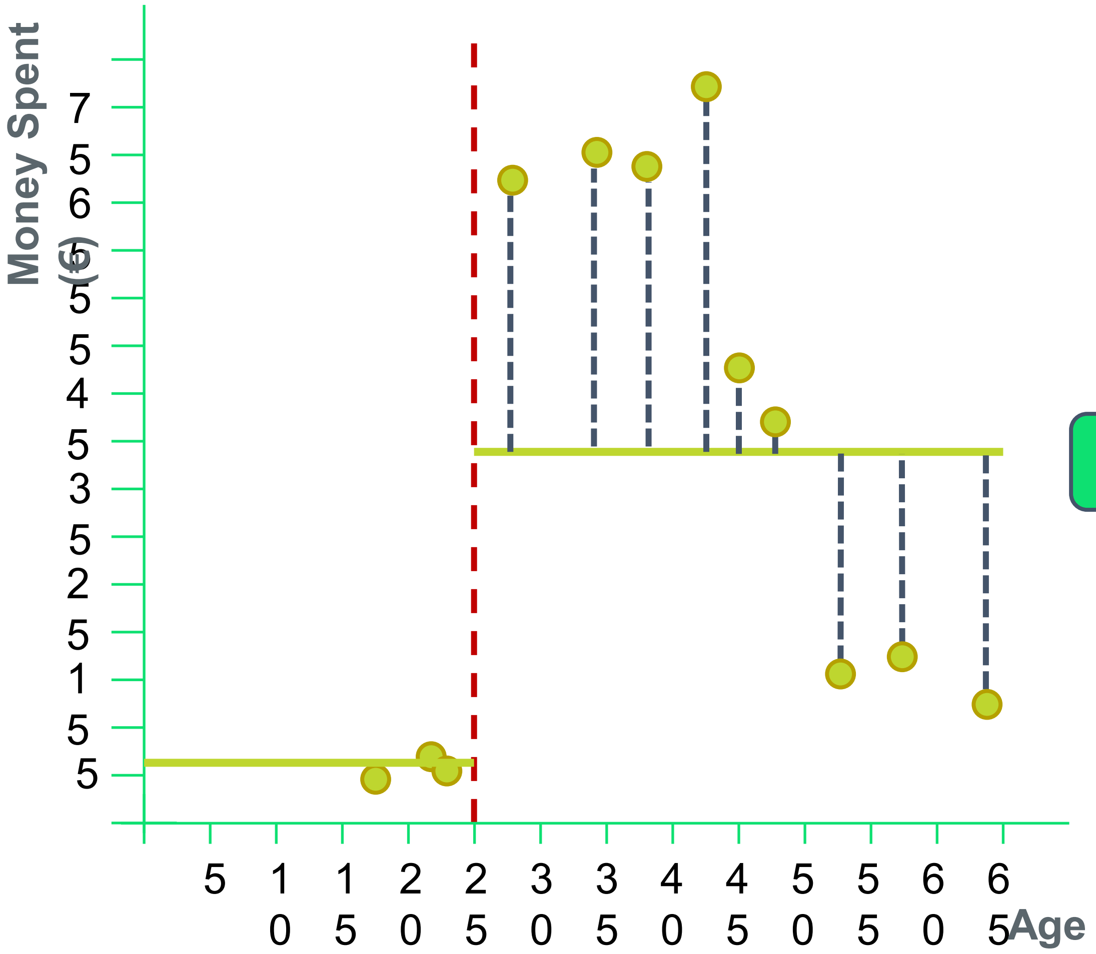


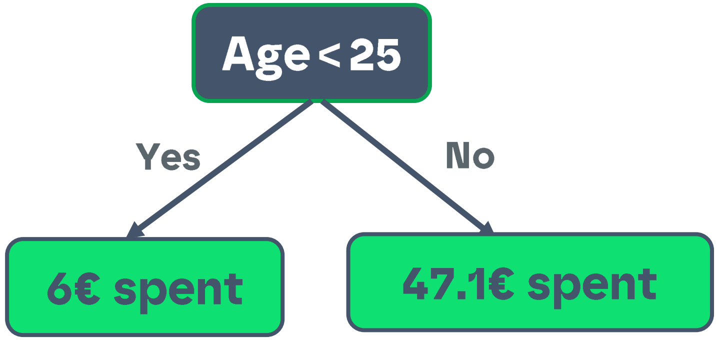
Age	Money Spent (y)	$\hat{y}$	MSE
18	5	6	1.0
22	7	6	
23	6	43	688.8
27	68	43	
34	72	43	
38	69	43	
43	77	43	
45	48	43	
48	42	43	
53	16	43	
57	18	43	
64	14	43	
		TOTAL MSE	689.8

MSE

2.1. Regression Trees

Using MSE

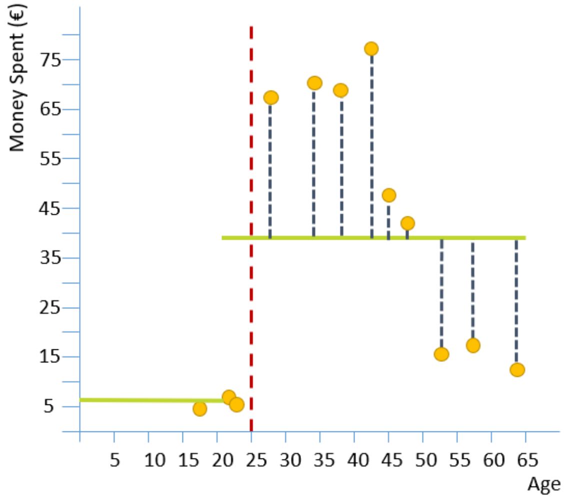


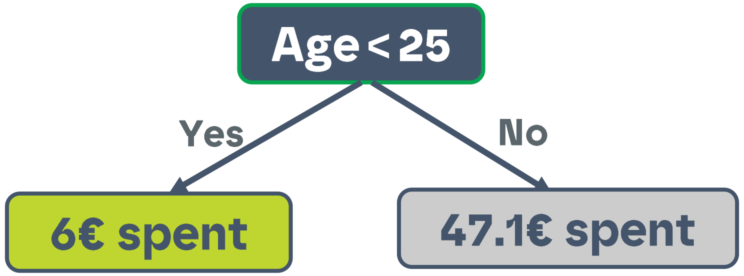


2.1. Regression Trees

Using MSE

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y})^2$$





Age	Money Spent (y)	$\hat{y}$	MSE
18	5	6	0.7
22	7	6	
23	6	6	
27	68	43	596.3
34	72	43	
38	69	43	
43	77	43	
45	48	43	
48	42	43	
53	16	43	
57	18	43	
64	14	43	
		TOTAL MSE	597.0

MSE

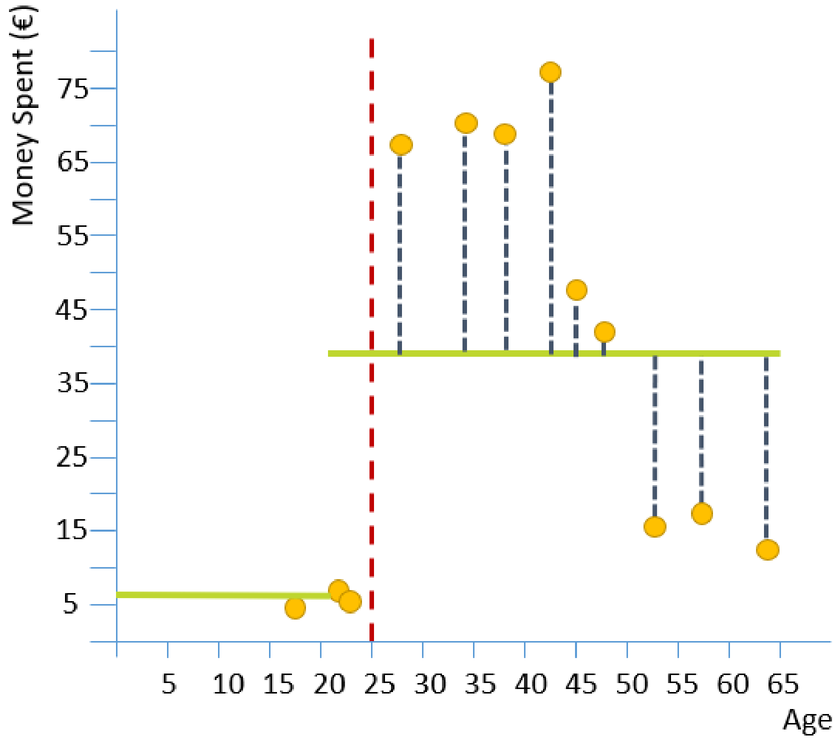
2.1. Regression Trees

Using MSE

Splitting criteria for age	TOTAL MSE
<20.0	733.3
<21.5	689.8
<25.0	597.0
< 30.5	1330.8
< 36.0	1558.1
< 40.5	1526.6
< 44.0	1265.0
< 46.5	1056.8
< 50.5	828.0
< 55.0	816.2
< 60.5	782.1

This is my first splitting criteria!

Age lower than 25



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MSE

2.1. Regression Trees

Using MSE

And my second splitting criteria? (MSE)

Age < 25

Yes:

Age	Money Spent (y)
18	5
22	7
23	6

No:

Age	Money Spent (y)
27	68
34	72
38	69
43	77
45	48
48	42
53	16
57	18
64	14

If you continue this process, you will end with one observation (since all values on money spent are different) in each leaf…

OVERFITTING!!

!The need to define stopping criterias!

MSE

2.1. Regression Trees

Using MSE

And my second splitting criteria? (MSE)

If I define, for example, the following stopping criteria:
- Minimum samples to split = 7

Age < 25

Yes

Only 3 samples!
I finish here

6€ spent

No

9 samples...
I need to check the MSE for each split

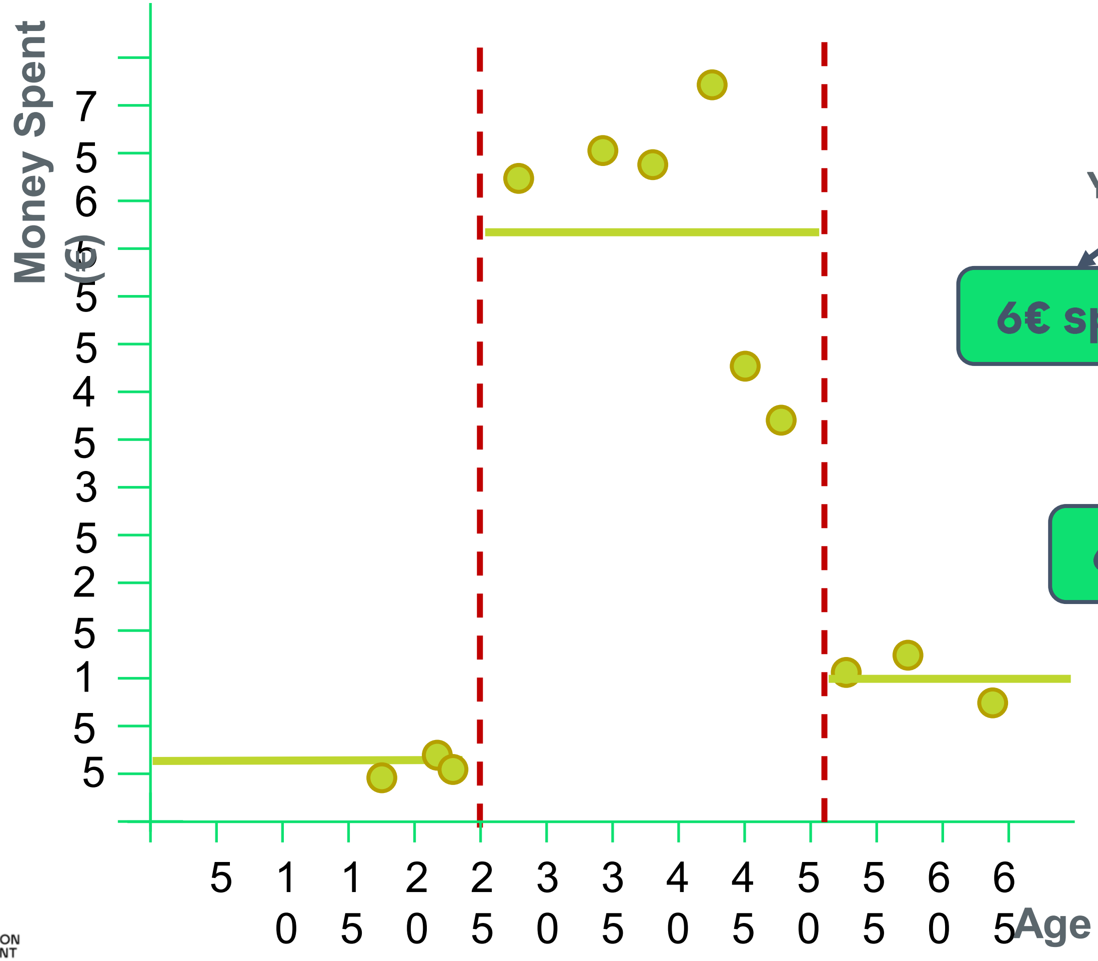
Splitting criteria	Total MSE
< 30.5	609.5
< 36.0	577.1
< 40.5	514.4
< 44.0	219.3
< 46.5	226.9
< 50.5	169.9
< 55.0	414.0
< 60.5	516.7

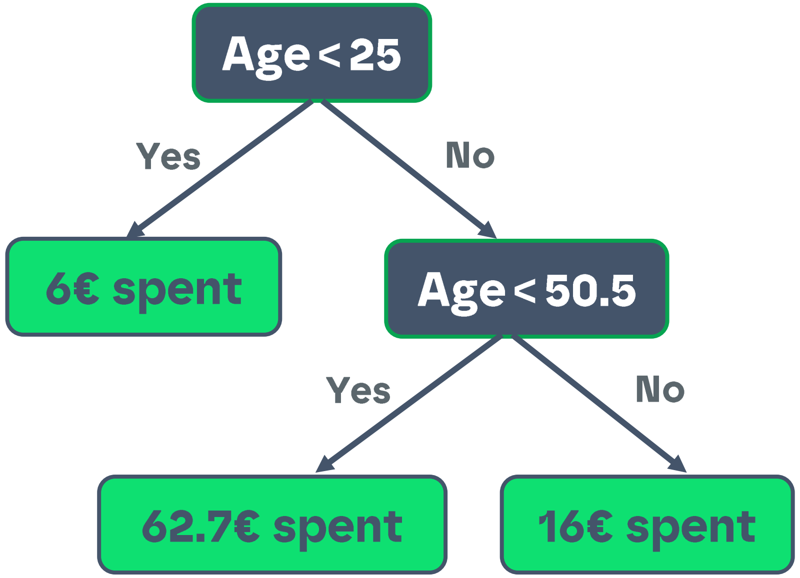
MSE

2.1. Regression Trees

Using MSE

My final tree (MSE)





MSE

2.1. Regression Trees

Using MSE

Using more than two predictors

Age	#Kids	#Months as customer	...	Money Spent
18	0	1	...	5
22	1	3	...	7
23	0	6	...	6
27	2	22	...	68
34	3	27	...	72
38	2	29	...	69
43	1	15	...	77
45	0	12	...	48
48	3	17	...	42
53	2	18	...	16
57	3	43	...	18
64	1	32	...	14

Just like before, we will try different thresholds for Age and calculate the MSE (or other measure) at each step, and pick the threshold that gives us the minimum MSE.

The best threshold becomes a candidate for the root

Age < 25

MSE

2.1. Regression Trees

Using MSE

Using more than two predictors

Age	#Kids	#Months as customer	...	Money Spent
18	0	1	...	5
22	1	3	...	7
23	0	6	...	6
27	2	22	...	68
34	3	27	...	72
38	2	29	...	69
43	1	15	...	77
45	0	12	...	48
48	3	17	...	42
53	2	18	...	16
57	3	43	...	18
64	1	32	...	14

Splitting criteria for #Kids	TOTAL MSE
<1	1155.8
<2	1301.1
<3	1321.6

MSE

2.1. Regression Trees

Using MSE

Using more than two predictors

Age	#Kids	#Months as customer	...	Money Spent
18	0	1	...	5
22	1	1	...	7
23	0	3	...	6
27	2	2	...	68
34	3	2	...	72
38	2	3	...	69
43	1	2	...	77
45	0	1	...	48
48	3	2	...	42
53	2	1	...	16
57	3	3	...	18
64	1	2	...	14

Splitting criteria for #Months	TOTAL MSE
<2	1056.7
<3	1501.3

MSE

2.1. Regression Trees

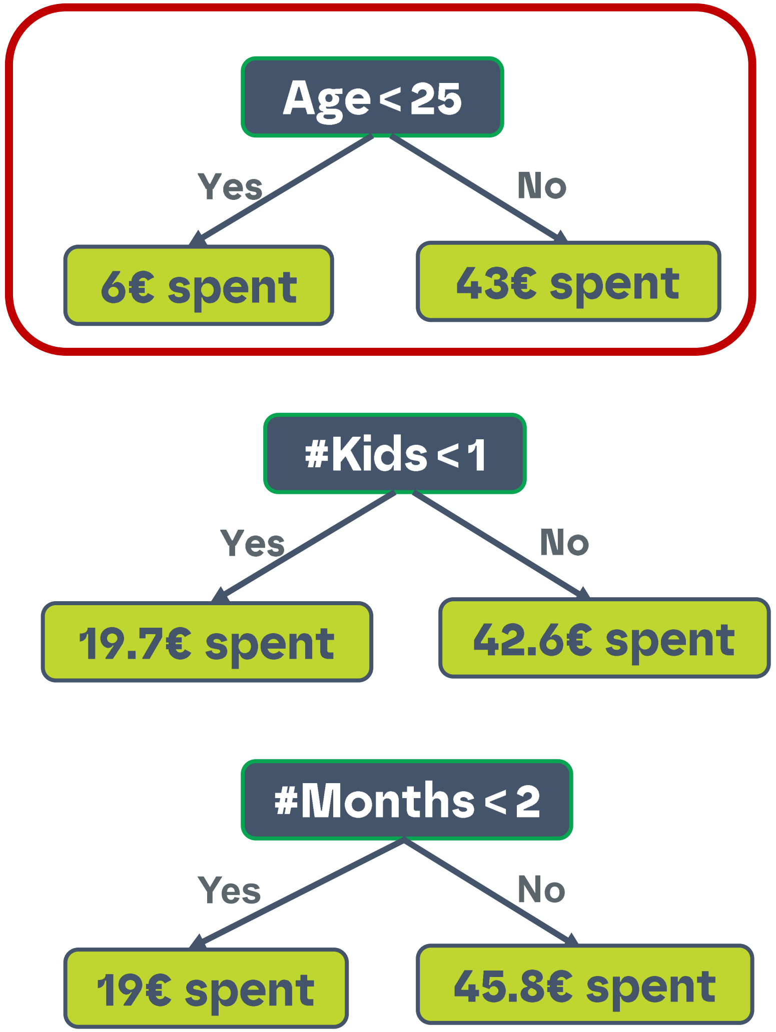
Using MSE

Using more than two predictors

Age	TOTAL MSE
<20.0	733.3
<21.5	689.8
<25.0	597.0
< 30.5	1330.8
< 36.0	1558.1
< 40.5	1526.6
< 44.0	1265.0
< 46.5	1056.8
< 50.5	828.0
< 55.0	816.2
< 60.5	782.1

# Kids	TOTAL MSE
<1	1155.8
<2	1301.1
<3	1321.6

#Months	TOTAL MSE
<2	1056.7
<3	1501.3



2. Regression Trees
Other measures

-Similarly to what we have on classification trees, where gini or entropy can be used to measure the quality of a split, in regression trees we have different alternatives also…

-MSE

-MAE

-Friedman MSE

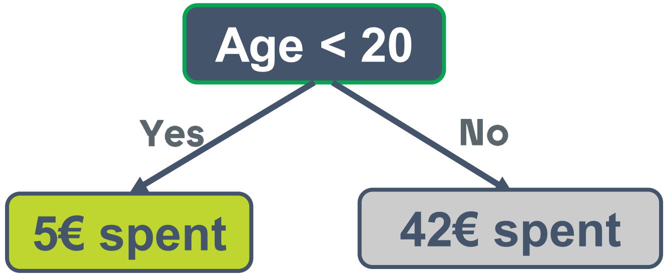
-Sum of residuals

-…

-In sklearn, you have the three first options

2.2. Regression Trees

Using MAE



MAE

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Age	Money Spent (y)
18	5
22	7
23	6
...	...
64	14

Median (y) = 5

$$MAE\ (Age < 20) = |5 - 5| = 0$$

Median (y)= 42

$$MAE\ (Age \geq 20) = \frac{1}{11} \times (|7 - 42| + |6 - 42| + \ldots + |14 - 42|) = 24.8$$

$$TOTAL\ MAE = 0 + 24.8 = 24.8$$

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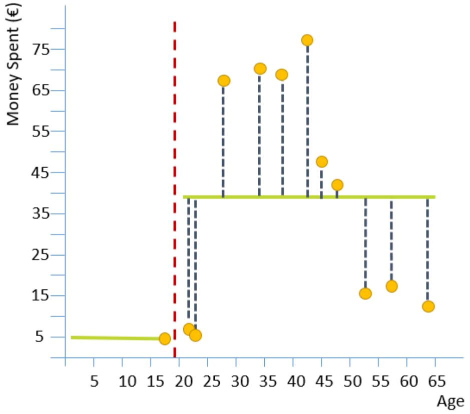
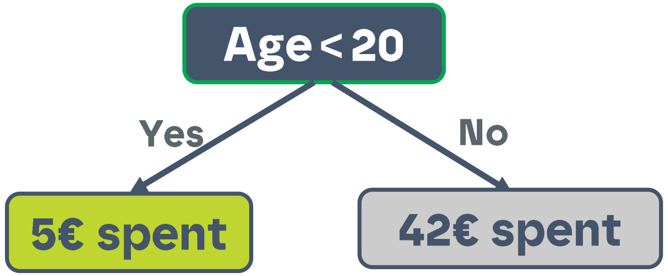
2.2. Regression Trees

Using MAE

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$





Age	Money Spent (y)	$\hat{y}$	MAE
18	5	5	0.0
22	7	39.7	24.8
23	6	39.7	
27	68	39.7	
34	72	39.7	
38	69	39.7	
43	77	39.7	
45	48	39.7	
48	42	39.7	
53	16	39.7	
57	18	39.7	
64	14	39.7	
		TOTAL MAE	24.8

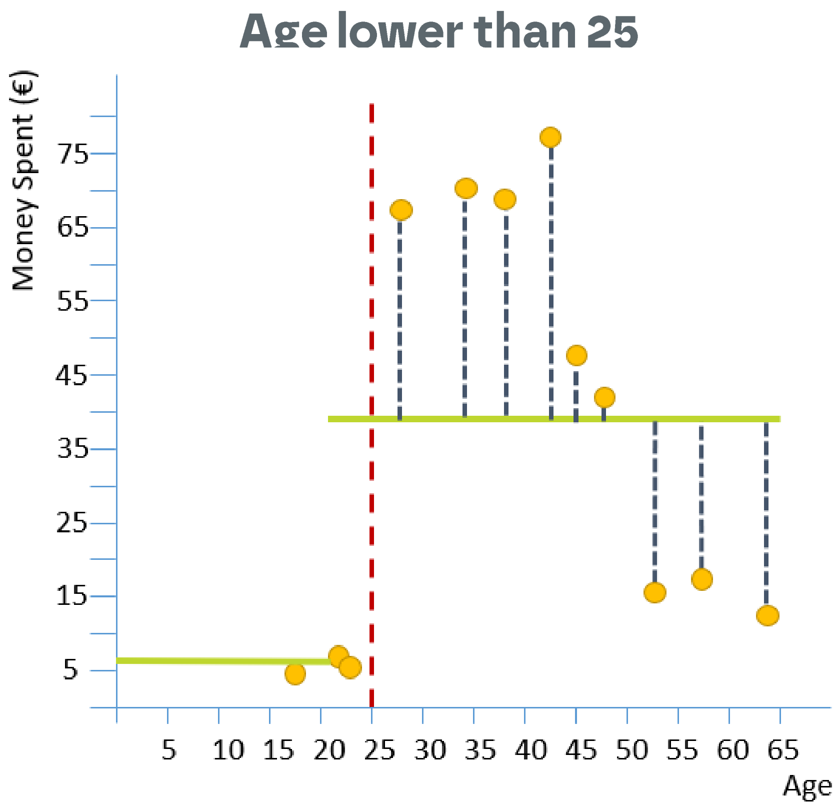
MAE

2.2. Regression Trees

Using MAE

Splitting criteria for age	TOTAL MAE
<20.0	24.8
<21.5	22.7
<25.0	18.9
< 30.5	23.2
< 36.0	26.8
< 40.5	57.7
< 44.0	80.9
< 46.5	70.9
< 50.5	60.0
< 55.0	60.2
< 60.5	59.3

This is my first splitting criteria!



2.2. Regression Trees

Using MAE

And my second splitting criteria? (MAE)

If I define, for example, the following stopping criteria:

- Minimum samples to split = 7

Age < 25

Yes

Only 3 samples!
I finish here

6€ spent

No

9 samples…
I need to check the MAE for each split

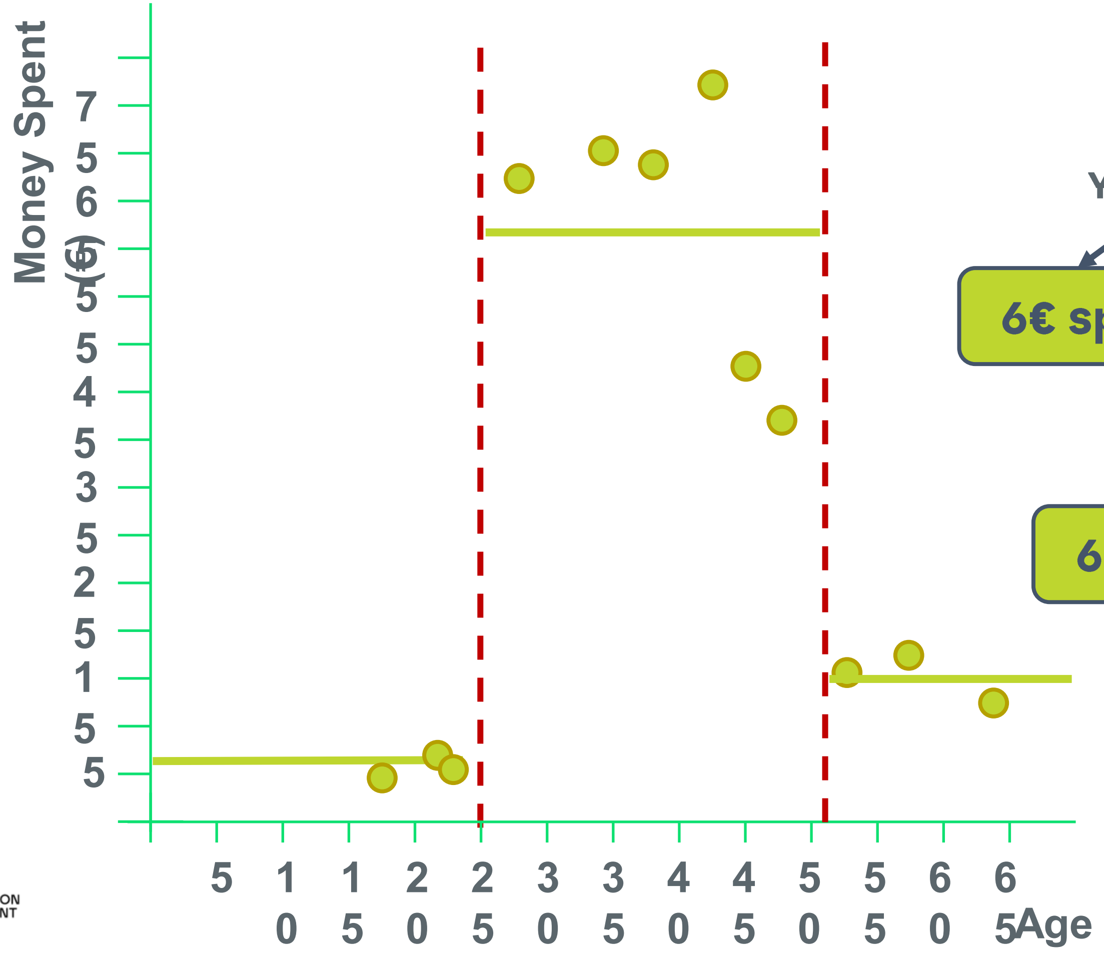
Splitting criteria	Total MAE
< 30.5	22.0
< 36.0	22.9
< 40.5	21.2
< 44.0	15.0
< 46.5	14.1
< 50.5	11.3
< 55.0	18.4
< 60.5	20.3

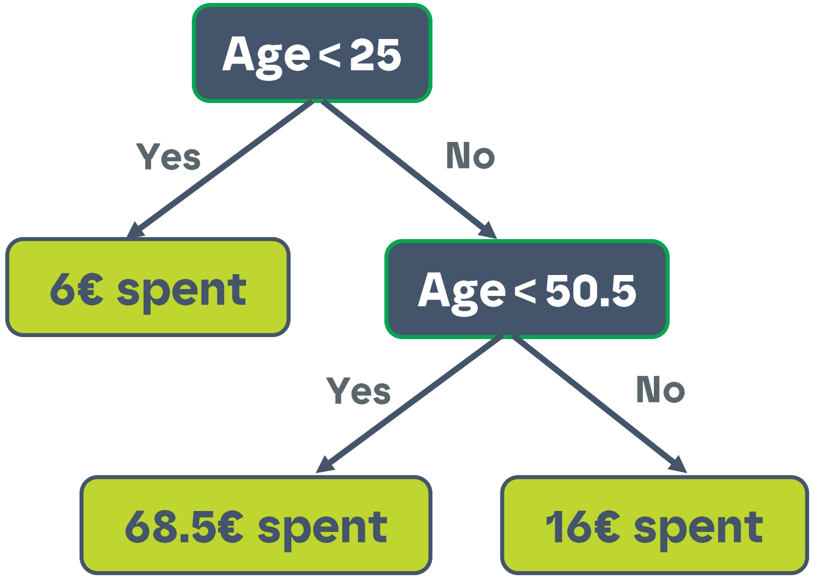
2.2. Regression Trees

My final tree (MAE)

Using MAE

MAE





03.

How to avoid overfitting in DT?

Two main approaches: prepruning and postpruning

3. How to avoid overfitting in DT?

Introduction

How to avoid overfitting in decision trees?

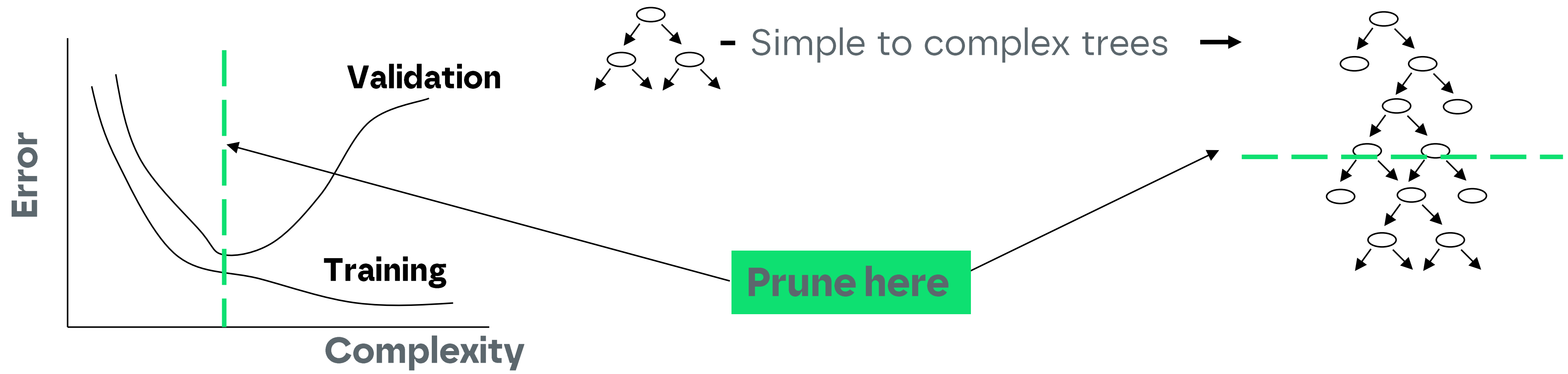
- An induced tree may overfit the training data
  - Too many branches, some may reflect anomalies due to noise or outliers
  - Poor accuracy for unseen samples

Two approaches to avoid overfitting

- ***Prepruning***:
  - do not split a node if this would result in the goodness measure falling below a threshold
  - Difficult to choose an appropriate threshold
- ***Postpruning***:
  - Remove branches from a “fully grown” tree—get a sequence of progressively pruned trees
  - Use a set of data different from the training data to decide which is the “best pruned tree”

3. How to avoid overfitting in DT?
Introduction

How to avoid overfitting in decision trees?



- **Training set** – used to develop the tree
- **Validation set** – used to access the generalization ability of the tree

References

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